

The Closure Challenge

A post-hoc RANS correction entry, its verification record,
and the field-inversion line behind it

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Status. Nothing has been submitted to the Closure Challenge. No account has been created and the benchmark steward has not been contacted. The entry described here (overall **0.056647**, round 5) is an internally measured position against the public leaderboard, **scored locally** through the benchmark’s own unmodified scorer at benchmark commit `deb91557`; it is not an official ranking, and where this document says *rank 1* it means rank 1 on that local scoring — a point estimate whose probability is **$P(\text{rank } 1) = 68\%$** , and eight cases cannot pin that tighter than 2–100% at 95%, with the lead over the rank-2 entry **not statistically decided** (Table 2). The submission package exists, is audited, and is one email away. It is **held at the owner’s discretion** — not withdrawn, not abandoned, not sent; the decision rests with Katie Kim.

Abstract

This report records what the Certonomous CFD laboratory has done in the Closure Challenge, a public benchmark for machine learning in turbulence modelling stewarded by Ryley McConkey (MIT). Our entry of record scores **0.056647** on the benchmark’s own unmodified scorer, which is the best overall number on the board as published at benchmark commit `deb91557` — **rank 1 of 5, scored locally**, 0.002878 below Reissmann, Fang & Sandberg’s published 0.059525, and 45.3% below the 0.1036 scored by submitting the supplied RANS fields unchanged. That rank is our own local scoring of an unsubmitted entry and is stated with that caveat everywhere it appears in this report, and now with its posterior beside it: **$P(\text{rank } 1) = 68\%$** over a 400,000-draw case-level bootstrap (an eight-case sample cannot pin that tighter than 2–100% at 95%), the lead over the rank-2 entry **not statistically decided** (paired $t = -0.495$, $n = 8$) and the lead over rank 3 likewise ($t = -0.953$); the two remaining comparisons are decided on method (Table 2). The entry is now *two* method classes rather than one, and round 5 is the reason: five of the eight predictions remain post-hoc velocity-field corrections, two of them withheld by a train-only *decline gate* that submits the untouched baseline where it predicts correction would do harm, while the three ducts are converged forward solves of an untrained QCR2000 constitutive term with nothing fitted to anything — adopted at Spalart’s published constant and shipped on all three ducts or none, under a rule frozen before the first validation iteration existed. We state the costs as plainly as the benefits: the two corrected hills still violate continuity at about 10% of the local velocity-gradient scale, where the ducts, now coming out of a solve, no longer do (27–63× cleaner than the fields they replace); the round-4 lead of 0.00003 on `AR_14_Ret_180` was priced as at-risk in writing before the run, was lost, and was not reverted; and two of the four leaderboard-best rows we now hold are the organisers’ own baseline rather than our model. The report also documents the parallel field-inversion line, which in the last week moved from a hard numerical blocker (`PetscConvergedReason: -9`, reproduced on two cases and five matrix orderings) to the laboratory’s first converged, finite-difference-verified closure-relevant adjoint gradient — and then to a negative result: the first inversion failed both of its pre-registered gates, and the post-run audit found the objective, not the method, at fault.

Provenance convention. Every quantitative claim below is followed by the repository artifact it comes from. Repository paths are relative to the `Certonomous` repository root and are cited

with the commit at which the cited text stands; external work is cited by DOI or arXiv identifier and was read from a copy held under `docs/papers/`. Appendix A is the full provenance index. Where a number is a bound rather than a measurement, or an inference of ours rather than a claim of the source’s, this report says so at the point of use.

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1 The challenge, and the metric it is scored on

1.1 What the benchmark is

The Closure Challenge is an open, ongoing benchmark task for machine learning in turbulence modelling. Its authoritative surface is the repository <https://github.com/rmcconke/closure-challenge-benchmark>, held locally at commit `deb91557`; the evaluation package `closure-challenge` is pinned at commit `1c4e22c8`; the accompanying preprint is arXiv:2603.28884 (McConkey, Buchanan, Smidt, Bodner, Dwight, Cinnella), DOI 10.48550/arXiv.2603.28884. The README states in its own words that it, not the paper, is the current source of information, and our records follow that ordering: no rule in this document is sourced to the preprint where the README states the same thing (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §1–§2, commit `15530f97`).

The task is to predict the velocity field on eight held-out test cases given a suggested training and validation split and a supplied CFD mesh. The eight scored cases are four parametric periodic hills (`alpha_15_13929_4048`, `alpha_15_13929_2024`, `alpha_05_4071_4048`, `alpha_05_4071_2024`), three rectangular ducts (`AR_1_Ret_360`, `AR_3_Ret_360`, `AR_14_Ret_180`) and the NASA wall-mounted hump (`NASA_2DWMH`).

Three properties of the rules matter to everything that follows.

1. **There is exactly one strict rule**, and it is about data, not method: it is “*strictly forbidden to train or validate on any data from the test cases*”, on pain of automatic withdrawal and a permanent note on the leaderboard. The preprint adds the permission the entry leans on, verified verbatim from the preprint’s own §2.1 on 2026-08-02: “*You can train on similar flows to the test cases.*”
2. **Everything else is unrestricted**. Input features, base turbulence model and data-assimilation technique are explicitly the submitter’s choice; the published train/validation split is labelled “(*suggested*)” and only the test column binds.
3. **There is no scoring-call limit, and there cannot be one**. The evaluation package ships the test ground truth locally and the README instructs submitters to preview their score; `evaluate_by_case()` is documented public API. Our six-calls-in-total ledger (§6.3) is therefore a self-imposed discipline, *stricter* than the benchmark requires, and must never be described as compliance with a rule that does not exist.

There is no eligibility clause of any kind in the benchmark README, the evaluation package README or the preprint — an absence of restriction, reported here as an absence rather than as a permission. No fee, registration, account or affiliation is required; submission is an email of a `test/` directory to the steward.

1.2 The metric, read from the evaluation code

The metric was read directly from `closure_challenge/eval.py` rather than from our own prose (`CLOSURE_METHODS_COMPARISON.md` §1.0, commit `ac2f37ee`). For a case with 1000 fixed evaluation points, `evaluate_individual_case()` computes a scaled mean absolute error of the *vector magnitude* of the velocity error,

$$s_{\text{case}} = \frac{\frac{1}{N} \sum_{i=1}^N \|\mathbf{U}_i^{\text{pred}} - \mathbf{U}_i^{\text{true}}\|_2}{\frac{1}{N} \sum_{i=1}^N \|\mathbf{U}_i^{\text{true}}\|_2}, \quad N = 1000, \quad (1)$$

and `score()` returns the plain, unweighted mean of the eight per-case values. Lower is better. Two consequences are worth carrying: the error is a vector-magnitude error and not component-wise, and because the mean is unweighted, an improvement of Δ on one case moves the overall

by $\Delta/8$ — so the two `alpha_15` cases, whose baseline errors of 0.1320 and 0.2049 dominate the floor, carry exactly the same $1/8$ weight as everything else.

All scoring in this project is done through the benchmark’s own unmodified scorer. The metric is never reimplemented for scoring purposes.

1.3 The public leaderboard, and where we stand

Table 1 gives the public board as published at benchmark commit `deb91557`, with our unsubmitted entry inserted at its measured position. All four published entrants re-score to their published values exactly on our harness, per-case and overall, which is the strongest available evidence that the harness is being driven correctly (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §7.5).

Table 1: The public leaderboard, with our entry of record at its internally measured position. We are unsubmitted; this is a local scoring, not an official ranking. Published ranks are the board’s own; ours is where our number falls in it. The position is worth $\mathbf{P(\text{rank } 1) = 68\%}$ — 2–100% at 95%, the width eight cases support — and the two rows immediately below ours are **not statistically decided** against it (Table 2).

Rank	Entry	Overall
—	Certonomous (unsubmitted, round 5, entry of record)	0.056647
1	Reissmann, Fang & Sandberg	0.0595
2	Wu & Zhang	0.0624
—	Certonomous (unsubmitted, round 4, superseded)	0.0654
—	Certonomous (unsubmitted, round 3, superseded)	0.0676
3	Liu, Wang, Zhao & Xiao	0.0737
4	Montoya, Oulghelou & Cinnella	0.0779
—	RANS-identity floor (submit the supplied fields unchanged)	0.1036

Rank 1, stated once with its caveat attached, and never stated without it. As of benchmark commit `deb91557`, scored on 2026-08-07 through the benchmark’s own unmodified scorer on the identical local path rounds 1–4 used, our round-5 entry’s 0.056647 is the best overall number on that board, 0.002878 below Reissmann’s published 0.059525. **This is a local scoring and not an official leaderboard placement.** Nothing has been submitted; no steward has scored us; if the steward’s own scoring differs from ours, the steward’s number is the number. Every appearance of “rank 1” below carries this sentence by reference (`CLOSURE_CHALLENGE_STATUS.md` §0f; `closure_challenge_round5_qcr.json`).

And a probability, because a rank is a point estimate. *Added 2026-08-10, under ladder V’s V8 clause: no rank claim ships without its probability and its not-decided pairs.* Resampling the eight cases — 400,000 case-level bootstrap draws over the per-case scores already on disk, no new scoring call — puts $\mathbf{P(\text{rank } 1) = 68\%}$ (67.6%), which eight cases cannot pin tighter than **2–100% at 95%** — the full ladder of intervals is unpacked immediately after Table 2, and the figure does not travel anywhere without one of them. The rest of the distribution is not a rounding error: rank 2 at 18.6% and rank 3 at 12.8%.

Read in the order of increasing honesty. The Monte Carlo interval (67.5–67.8%) measures only that the resampling ran long enough. Leave-one-case-out across eight refits gives 38–91%, and a double bootstrap gives 26–94% at the 68% level and 2–100% at 95%. **Eight cases cannot resolve a probability of rank better than “somewhere between a coin flip and near certain”**, and that sentence is the most useful line in the underlying analysis rather than a defect of it. A second source of doubt sits outside the method entirely: the seed-spread bound on the three seed-dependent cases is 0.0024 against a rank-1 margin of 0.002878, and loading it adversely moves $\mathbf{P(\text{rank } 1)}$ from 68% to **52%** (favourably, to 80.5%). The honest sentence this

Table 2: The rank claim with its posterior attached (`campaign/PROBABILITY_OF_RANK_2026-08-10.md`, 400,000 case-level bootstrap resamples; the figure is publishable and **may not be stated without its interval** — see the caution below). t is the paired per-case difference statistic, $n = 8$.

comparison	P(we lead)	paired t	status
Reissmann, Fang & Sandberg (margin 0.002878)	69.4%	−0.495	not statistically decided
Wu & Zhang (margin 0.005765)	84.7%	−0.953	not statistically decided
Liu, Wang, Zhao & Xiao (margin 0.017040)	98.7%	—	decided on method
Montoya, Oulghelou & Cinnella (margin 0.021216)	99.8%	—	decided on method

supports is: *we are probably ahead, we cannot show that we are, and the two largest sources of doubt — which eight cases, which seed — are both outside our method.*

Against Reissmann specifically: we win four of the eight cases and lose four, and the per-case dispersion is five times the margin. A 69.4% probability of leading is what “we might well be ahead” looks like written down, and it is **not** a claim that our method is better than theirs. Two per-case comparisons must never be quoted as wins or losses at all: `AR_1_Ret_360` (ours 0.045470 against a published 0.0455 — a gap of 0.00003, below the precision the board publishes to) and `AR_3_Ret_360` (0.039982 against 0.0399). Both are ties at the board’s own precision.

Two further cautions attach to that table and are repeated wherever it appears.

- **The 0.1036 floor is our own number.** The script that produces it (`scripts/rans_identity_baseline.py` in the benchmark clone) is untracked — it was written by this laboratory, not shipped by the benchmark, and the README quotes no RANS-identity baseline anywhere. It reproduces exactly and is honestly derived, but any wording implying the floor is the benchmark’s own published figure is false (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §7.2).
- **The rank is a point estimate, and the probability travels with it — never without its interval.** $P(\text{rank } 1) = 68\%$, which eight cases cannot pin tighter than **2–100% at 95%**; the leads over Reissmann and over Wu & Zhang are **not statistically decided** (Table 2). *Until 2026-08-10 this item read differently: it marked that figure internal to this laboratory — not to be quoted outside it, attached to any package, or cited in any submission-facing document. That restriction was withdrawn, and the reasoning is recorded here rather than deleted, because a reader who met the old rule is owed the reason it went.* The cold-reproduction pass recomputed the figure as **0.674** from public data alone — the benchmark’s own published board and our eight per-case scores — in about a minute, with no access to the internal document. A number an outsider reproduces that easily is not protected by being withheld; it only looks concealed, and it looks that way to exactly the reader the disclosure is meant to convince. Publishing “rank 1” while withholding how little that rank survives resampling also inverts this laboratory’s own standard. So the figure now travels with the entry, and a single prohibition replaces the old internal/external split: **no surface may state it without the interval** — a bare 68% sounds settled, and eight cases do not support settled (`campaign/PROBABILITY_OF_RANK_2026-08-10.md` head banner, commit `b032bbae`; `campaign/LADDER_V_PASS3_COLD_2026-08-11.md` §3 (W2), commit `9477a2ed`; `campaign/LADDER_V_TRIPLE_VERIFICATION.md` V8 amendment, commit `5bffb476`).
- **Position claims go stale.** The challenge is ongoing with no deadline; the board has been unchanged since 2026-05-04 but can change without notice. Every position claim in this document is dated 7 August 2026. A rank-1 claim goes stale faster than a rank-3 claim did, because it is the row a new entrant most easily displaces.

2 Our approach

2.1 Method class, stated at its real strength

Our correction is a **post-hoc velocity-field correction**. The learning target is $\delta\mathbf{U} = \mathbf{U}_{\text{LES}} - \mathbf{U}_{\text{RANS}}$ per mesh cell, and the learned map is applied once to an already-converged RANS field as a final post-processing step. **Nothing re-enters the governing equations and nothing is re-solved with the correction folded in.** In the Duraisamy–Iaccarino–Xiao taxonomy — *Turbulence Modeling in the Age of Data*, *Annu. Rev. Fluid Mech.* 51 (2019) 357–377, cited at its honest tier, metadata only: the review has not been read in full by this laboratory, so “correct the answer” is our own paraphrase of the family via `docs/research/CLOSURE_METHODS.md`, not the review’s wording — this is structurally distinct from FIML, TBNN, SpARtA and eigenvalue perturbation, all of which correct something inside the equations before re-solving (`CLOSURE_CHALLENGE_STATUS.md` §4, commit `b9cb56a2`).

Update, 2026-08-07 — that description is now true of five predictions and false of three. Round 5 replaced the duct correction with a converged forward solve of an untrained QCR2000 constitutive term (§3.2). The entry of record is therefore a **hybrid**, and this report says so at the front rather than letting the old one-line self-description stand: `alpha_15_13929_4048`, `alpha_15_13929_2024` and `NASA_2DWMH` are post-hoc corrected velocity fields; `alpha_05_4071_4048` and `alpha_05_4071_2024` are the organisers’ own unmodified solve, withheld by the gate; and `AR_1_Ret_360`, `AR_3_Ret_360` and `AR_14_Ret_180` are our own SIMPLE solves of a modified constitutive relation, in which the correction most certainly does re-enter the governing equations. **The claim “unique among the five in not re-solving” has been retired, not softened:** on three of eight cases we now do what everyone above us was doing, and §3.2 records that we did it because the measurement said the post-hoc class could not reach the duct structure at all. Everything below that describes the post-hoc class describes five of our eight predictions, and this report marks which is which wherever the distinction bites.

Correction, 2026-08-05 — the family we are in now carries its citation. The passage above previously placed the method in a taxonomy, named four families we are not, and cited no prior work for the family we are in — origination by omission, named as such by the method-priority review (`CLOSURE_METHOD_PRIORITY_REVIEW.md` §4.1, commit `d84b649f`). The owed citation is paid here, at read-in-full tier: learned local-error correction of a cheap CFD solve, applied once without re-solving, is established — Hanna, Dinh, Youngblood & Bolotnov, *Coarse-Grid Computational Fluid Dynamic (CG-CFD) Error Prediction using Machine Learning*, arXiv:1710.09105 (2017); journal version *Prog. Nucl. Energy* 118 (2019) 103140, DOI 10.1016/j.pnucene.2019.103140 (journal record: metadata only). The arXiv full text was fetched and read on 2026-08-05 and is held at `docs/papers/hanna_dinh_youngblood_bolotnov_1710.09105.pdf`. Their surrogate predicts the cheap solve’s local error from the cheap solve’s own local features and adds it back to the variables of interest, nothing re-solved — our architecture with the error source swapped: their cheap solve is a coarse-grid *no-model* Navier–Stokes solve of a lid-driven cubic cavity and the corrected error is grid-coarsening (discretisation) error, tested across unseen Reynolds numbers and grid sizes; ours is a converged $k\text{--}\omega$ SST solve and the error is turbulence-closure error on separated and secondary flows. Two deltas from the full read are worth carrying beyond the review’s abstract-tier characterisation: their 37 features are the cell Reynolds number plus scaled first and second velocity derivatives — grid-resolution features, not physics invariants — and their own open-issues section names the exact cost our G2 audit later measured on ourselves, that the velocity components “are corrected separately, without enforcing conservation and Galilean invariance”. The founding CFD paper of the class states the class’s physicality gap; it does not soften our measured instance of it (§6.5). The statistical ancestor is model-discrepancy calibration — Kennedy & O’Hagan, *J. R. Statist. Soc. B* 63(3)

(2001) 425–464, cited as lineage at metadata-only tier: not read in full by this laboratory, and no sentence of it is quoted. What survives, stated in the same breath: within the data-driven RANS-closure literature swept by the review (27 recorded searches, 13 counted negatives), every located correction that produces a velocity field re-enters the equations and re-solves; no other post-hoc *velocity-field* correction was found, so the application to turbulence closure is ours even though the class is not.

The published argument for the target choice. One prior argues *for* this method class rather than merely tolerating it: Wu, Xiao, Sun & Wang, *Reynolds-averaged Navier–Stokes equations with explicit data-driven Reynolds stress closure can be ill-conditioned*, *J. Fluid Mech.* 869 (2019) 553–586, arXiv:1803.05581 (abstract read 2026-08-05). Propagating a learned Reynolds-stress correction through the RANS equations is a published conditioning hazard — per the arXiv record’s own summary, stress errors below 0.5% can amplify to velocity errors up to 35%; those figures are quoted at the record’s tier and were not read off the paper’s tables — and every re-solving entrant on this board must survive it, where correcting the velocity field directly sidesteps it. This citation defends the choice of target as principled rather than merely cheap; it says nothing about the continuity cost measured in §6.5 and must not be used to soften it.

Two model objects stood behind the six corrected predictions of round 4, and one of them — the periodic-hill model, behind three of them — still ships in the entry of record (CLOSURE_METHODS_COMPARISON.md §1.1). Variant D is described here because it is the model round 5 displaced and because §3.2 is unreadable without it:

- **The periodic-hill (PH) model** — three HistGradientBoostingRegressors, one per velocity component, `max_iter=300`, `max_depth=6`, `learning_rate=0.05`, `l2_regularization=1.0`, `random_state=0`, on seven per-cell features: Pope’s five scalar invariants of the normalised strain and rotation tensors (`I1_S2`, `I2_W2`, `I3_S3`, `I4_W2S`, `I5_W2S2`; Pope, *J. Fluid Mech.* 72 (1975) 331–340), a wall-distance turbulent Reynolds number `Re_y`, and a turbulent-to-mean kinetic energy ratio `tke_ratio`. Trained on 21 periodic-hill cases, 327,600 cells.
- **The duct model, “Variant D”** — the same architecture on eight features (adding a Reynolds-invariant $d/d_{\max}$) with a velocity-scale-normalised target, $(\mathbf{U}_{\text{LES}} - \mathbf{U}_{\text{RANS}})/|\overline{\mathbf{U}_{\text{RANS}}}|$ per case, trained on the four suggested training ducts, 41,971 cells. **Superseded on 2026-08-07**: it appears in no round-5 prediction.
- **The duct model of record, kOmegaSSTQCR** — and it is not a model object in the machine-learning sense at all. It is a ~ 180 -line OpenFOAM library deriving from stock `kOmegaSST` and overriding `divDevRhoReff`, implementing Spalart’s QCR2000 quadratic constitutive relation $\tau_{ij} = \tau_{ij}^l - c_r(O_{ik}\tau_{kj}^l - \tau_{ik}^l O_{kj})$ at $c_r = 0.3$, the published constant, **untrained**; k and ω transport inherited unchanged. **It has no training set, no fitted parameter, no feature vector and therefore no leakage surface.** Its provenance and its falsifier record are §4.5 (`campaign/W3_QCR_DUCT_FALSIFIER.md`; `sdk/openfoam/qcr/kOmegaSSTQCR/`, library commit 303247bb).

Round 2 required building a feature pipeline the benchmark does not ship for the duct and CBFS families: Green–Gauss $\nabla \mathbf{U}$ reconstructed from mesh, RANS $\mathbf{U}$ and boundary conditions, and wall distance from the nearest wall-patch face centre. It was validated *only* against shipped fields on periodic-hill *training* cases: cell centres and volumes to $\sim 1\text{e-}13$ and $\sim 1\text{e-}22$ absolute, $\nabla \mathbf{U}$ to a pooled relative Frobenius error of 0.0012–0.0029 with Pearson $r = 0.9997$ –1.0000 on three different training cases, and wall distance to $r = 1.0000$ with mean relative error below 0.1% (CLOSURE_CHALLENGE_STATUS.md §4).

2.2 The do-no-harm gate

The distinguishing component of the entry is a **case-level decline gate**. It is a three-feature ridge regression (top-3 screened features `p90_I4_W2S`, `frac_backflow`, `p90_I3_S3`; RidgeCV $\alpha = 0.7499$) that predicts, from RANS-derived quantities alone, the scaled MAE the *uncorrected baseline* will achieve on a case. If the predicted baseline error falls below a threshold of 0.1263 — the leave-one-out median over the training cases — the correction is withheld and the unmodified RANS field *is* the submission for that case (`closure_challenge_C1_error_estimator.md`, commit 5719374e).

Every parameter of the gate — feature screening, ridge α , standardisation, final coefficients and threshold — was computed from the 21 periodic-hill training cases and nothing else, and was validated on the four suggested validation cases (AUC 1.0, 4 of 4 correct) before it was ever pointed at a test case. When it was applied, the refit reproduced $\alpha = 0.7499$ and threshold 0.1263 byte-for-byte from train-only data as a precondition of the run.

Two honest qualifications belong beside it:

- **Its statistical base is thin, and this was disclosed before it was useful.** AUC 1.0 on $n = 4$ with a single positive is approximately a one-in-four outcome by luck; the gate has made eight binary decisions in total. Widening the validation base to a 25-case leave-one-out is a filed proposal (G5, §6.4).
- **It is not novel, and the nearest prior art is by the challenge’s own authors.** Two established things, and they are not the same thing — the list below was one undifferentiated group until 2026-08-05, when the method-priority review (`CLOSURE_METHOD_PRIORITY_REVIEW.md` §4.2, commit d84b649f) found that grouping credited two of the four papers with a mechanism they did not report. **Identifying** where the baseline is unreliable, from RANS-only inputs, is established: Ling & Templeton, *Phys. Fluids* 27, 085103 (2015) — whose abstract, read at the OSTI record, classifies RANS results point by point as high or low uncertainty and controls no correction — and Wu, Wang, Xiao & Ling, *Flow Turbul. Combust.* 99, 25 (2017), DOI 10.1007/s10494-017-9807-0, an *a priori* confidence measure. **Using such a classifier to control where a data-driven correction is fitted and applied** is established separately: Steiner, Dwight & Viré, *Flow Turbul. Combust.* (2022), DOI 10.1007/s10494-022-00346-6; and Buchanan, Lăcătuș, West & Dwight, *Computers and Fluids* 305, 106899 (2025), DOI 10.1016/j.compfluid.2025.106899, arXiv:2504.06758 — whose first and last authors are co-authors of the Closure Challenge paper itself. Read in full, their RITA classifier selects *regions inside a case* from local ratios of the baseline’s own k -equation terms. What survives of our distinction, and all we claim, is: our gate acts on a *whole case*; it predicts the *baseline’s error* rather than distribution shift; and its “off” state means the uncorrected field *is* the submission (`CLOSURE_CHALLENGE_PRIOR_ART.md` §2.4, §7.4).
- **The nearest miss, quoted rather than characterised** (*added 2026-08-05, method-priority review §2.2*). Ji, Luo, Zhou & Zhao, *Progressive Mixture-of-Experts with autoencoder routing for continual RANS turbulence modelling*, arXiv:2601.09305 (2026), **read in full** 2026-08-05, computes exactly our quantity — a whole-flow confidence from the baseline field alone, “the relative frequency of recognised points” — and then does something else with it: “if p_K falls below T_{accept} , the flow is deemed unknown, which triggers the continual learning process.” Its low-confidence branch *expands*, spawning an expert. There is no branch that submits the uncorrected baseline. That is the single closest published approach to our gate and it stops one step short — which is the strongest available support for our narrow claim, and it is narrow.

That last reading carried a constraint we wrote down before anyone felt the pull of it: Buchanan et al. train on the NASA wall-mounted hump, which is one of our eight scored test

cases. Their published model coefficients are therefore **off limits** to this entry — borrowing them or calibrating anything of ours against that model’s hump behaviour would make our submission indirectly trained on a test case.

2.3 Pre-registration

Every model choice that selects between candidate prediction sets is frozen in a committed document *before* the prediction set it selects is scored. The NASA hump’s model choice was pre-registered before the round-2 scoring call; Variant D was pre-registered on the benchmark’s own suggested duct validation case `AR_7_Ret_180` before any test score for it existed (`closure_challenge_duct_reynolds_transfer.json`, commit `d0992da2`); the `alpha_05` regime candidate wrote down its accept/reject rule, then *failed its own pre-declared gate and was not scored* (§6.6); and round 5 froze the rule that decides where its model is applied *before the first validation iteration existed*, then accepted its own result mechanically (§6.2).

Pre-registration that only ever ratifies is decoration. The `alpha_05` no-go is the evidence that this one can say no; round 5 is the evidence that it can also say yes without anybody’s judgement being exercised after the number existed. Both halves are needed and the laboratory now has both.

Whose discipline this is, added 2026-08-05. Neither half of this practice is ours, and the method-priority review found this the element our records named the fewest priors for (`CLOSURE_METHOD_PRIORITY_REVIEW.md` §2.3, commit `d84b649f`). Freezing an analysis plan before results is an established practice in machine learning with its own venue: the **NeurIPS 2020 Workshop on Pre-registration in Machine Learning**, PMLR volume 148, held 11 December 2020 (editors Bertinetto, Henriques, Albanie, Paganini, Varol), with a second edition as PMLR volume 181 (NeurIPS 2021) — cited at metadata-only tier from the publisher’s volume pages; no paper from either volume has been read here. And the problem our scoring-call ledger (§6.3) addresses has a canonical paper: Blum & Hardt, *The Ladder: A Reliable Leaderboard for Machine Learning Competitions*, arXiv:1502.04585, ICML 2015, PMLR 37 (abstract read 2026-08-05). Their abstract prices our answer for us: repeated evaluation against a holdout lets participants overfit it, and existing approaches “resort to poorly understood heuristics such as limiting the bit precision of answers and the rate of re-submission.” **Our scoring-call ledger is rate limiting — one of those heuristics, self-imposed — and it should be read as harm reduction rather than a guarantee.** Blum & Hardt build a principled alternative and we do not implement it. What we did not find in that literature is an entrant imposing either discipline on itself, unasked, on a benchmark that imposes none, and publishing the ledger; that is a disclosure practice, not a method, and it is claimed as one.

2.4 The costs of this method class, stated before the results

Table 3 compares the five entries dimension by dimension, from each entrant’s own documents (`CLOSURE_METHODS_COMPARISON.md` §3.1). Read the top two rows first, because they are the ones that cost us.

The first three rows were the honest headline of this report when every one of our predictions was post-hoc, and half of that headline still stands. On the five cases where we correct a converged field, the submitted field is a cell-wise machine-learning output added to a solenoidal RANS solution and it does not satisfy continuity by construction. No realizability question arises — we predict velocity, not Reynolds stresses, so that column is not-applicable rather than passed — but $\nabla \cdot \mathbf{U}$ is a real property of a submitted field, and it was measured against a threshold fixed before the run with a finding material and against us (§6.5). **The other half of the headline was answered by changing the method rather than by arguing with the measurement:** the three duct predictions now come out of a converged SIMPLE solve, whose

Table 3: The five entries, dimension by dimension, with our column updated to round 5. “Absent from their documents” means absent from the sources read — two submission documents, one notebook plus info file, and two full papers — and is stated as an absence in those sources, not as a claim about any laboratory’s unpublished practice.

Dimension	Certonomous	Reissmann	Wu & Zhang	Liu et al.	Montoya et al.
Corrected object	velocity field, post-hoc (5 cases); untrained QCR2000 in the constitutive relation (3 ducts)	anisotropy + SST coefficients	β on ω -destruction + fixed QCR	TBNN anisotropy + k - ω coefficients, coupled	blended tensor-basis + R_k
Re-solved with the correction	on 3 of 8, from round 5 ; no on the other 5	yes (inferred)	yes	yes	yes
Conservation satisfied by the submitted field	ducts yes, by construction (round 5); hills no — measured (§6.5)	yes, by construction	yes	yes	yes
Trains on a scored case	no — enforced by executable asserts	no	no	yes, per their own paper (entry-level status unverifiable)	no
Case-level do-no-harm gate	yes	absent from their documents	region shield f_d , hand-designed	absent from their documents	baseline-as-expert local fallback
Pre-registration before scoring	yes	absent	absent	absent	absent
Scoring-call ledger	yes, 6 calls, self-imposed	none visible	none visible	none visible	none visible

pressure equation enforces continuity on the face fluxes, and their measured RMS $\nabla \cdot \mathbf{U}$ fell by 27–63 $\times$ against the fields they replace (§3.2). Of the two costs this section stated before the results, one is retired on three cases and one is unchanged on five.

The compensating claims stay narrow. Among the five entries on this board, ours is the only one that will decline to correct a whole case and submit the baseline instead; the only one that freezes its choices in advance of scoring; and, now, the only one whose duct predictions are produced by a term with no fitted parameter anywhere in it.

3 Results, round by round

3.1 The arc

Round 1 — 0.0869. The PH model, trained on 21 periodic-hill cases and validated on 4, applied to the four periodic-hill test cases only. All four non-PH test cases were passed through as unmodified raw RANS, identical to the floor, because the benchmark does not ship $\nabla \mathbf{U}$ and wall distance in solved time directories for the duct and CBFS families and reconstructing them was out of scope for that round.

Round 2 — 0.0741 (Δ −0.0128). The mesh-reconstruction pipeline of §2.1 unlocked the re-

maining four cases. A new duct-specific model was trained on `AR_1/3/5/10_Ret_180` and validated on the held-out `AR_7_Ret_180`; `NASA_2DWMH` was corrected with the existing PH model, a pre-registered choice made before test scoring. The hump regressed slightly, from 0.0621 to 0.0632, and was reported rather than reverted.

Round 3 — 0.0676 (Δ -0.0065). The C1 decline gate, built and validated in a prior session on non-test data only, was evaluated on the four periodic-hill test cases’ RANS-derived features for the first time. It declined both `alpha_05` cases (predicted baseline 0.0420 and 0.0034, both below the 0.1263 threshold) and kept the correction on both `alpha_15` cases (0.1503 and 0.1525, both above) — 4 of 4 correct on cases it had never been evaluated on, matching its validation performance exactly. The movement, -0.0065, matched the -0.0066 predicted by the round-2 error decomposition almost exactly. Compute: 83 s wall on a two-core cap, peak RSS 356 MB.

Round 4 — 0.0654 (Δ -0.0022). A duct-only change. Variant D replaced the round-2/3 duct model on all three ducts. The five non-duct predictions are *byte-identical* to round 3 — copied, not regenerated, and SHA-256-verified against the entry manifest — so the entire delta is attributable to the duct family and the gate’s behaviour is untouched.

Round 5 — 0.056647 (Δ -0.0088), the entry of record. A duct-only change again, and the largest single movement since round 2. The untrained QCR2000 term replaced Variant D on all three ducts, by forward solve rather than by correction; the same five non-duct predictions carried through byte-identical for a second round. The delta is again wholly the duct family, and it is the round in which the laboratory stopped trying to predict a flow structure and started solving for it (§3.2). It is also the round that put the entry at the top of the local board — at 68% probability, an eight-case figure no tighter than 2–100% at 95%, with the lead over the rank-2 entry **not statistically decided** (Table 2).

Round 4’s diagnosis was pre-registered and its sign pattern was predicted before it was measured. The published duct diagnosis had been *wrong*: `I3_S3` and `I4_W2S` are indeed algebraically zero on every duct, but they are equally zero on `AR_14_Ret_180`, the duct we score best on, so they discriminate nothing. What discriminates is that `Re_y` reaches $1.85\times$ and $2.07\times$ its trained maximum on the two ducts we were losing, against $0.90\times$ on the one we win — and a gradient-boosted tree cannot extrapolate. The pre-registration predicted improvement concentrated on the two extrapolating cases and little or none elsewhere. Measured: -0.0108, -0.0087, and +0.0022. The sign pattern matches on all three (`CLOSURE_CHALLENGE_STATUS.md` §0e).

3.2 Round 5: the untrained term takes the ducts

Round 5 changed three predictions and nothing else. On `AR_1_Ret_360`, `AR_3_Ret_360` and `AR_14_Ret_180` the round-4 machine-learned δU was discarded and the case was re-solved from the shipped `0/` fields on the shipped mesh, with the shipped schemes and `fvSolution` untouched, under `kOmegaSSTQCR` at $c_r = 0.3$. **Nothing in those three predictions was fitted to anything.** The other five CSVs were copied from the round-4 submission and hash-asserted against its manifest inside the producing script, so the whole movement is again the duct family (`campaign/R5_PREREGISTRATION.md` §3, commit `e865076b`).

Every quantity that could select a model was fixed before the number existed, and the mechanics of that are §6.2. The result of applying them (`closure_challenge_round5_qcr.json`; `CLOSURE_CHALLENGE_STATUS.md` §0f):

Case	Round 4	Round 5	Δ
AR_1_Ret_360	0.0811	0.045470	-0.0356
AR_3_Ret_360	0.0775	0.039982	-0.0375
AR_14_Ret_180	0.0325	0.035339	+0.0029
Overall	0.065431	0.056647	-0.0088

The five unchanged cases scored identically to round 4, as their byte-identity requires — which is also the cheapest available check that the harness was driven the same way twice. That 0.056647 is the number every rank claim in this report rests on, and its posterior is Table 2: **P(rank 1) = 68%**, an estimate eight cases pin no tighter than **2–100% at 95%**, with the two nearest comparisons **not statistically decided**.

Three things about that table are worth stating in the order the record forces them.

1. **The improvement landed where the pre-registered mechanism said it would, on a term with nothing fitted to anything.** Round 4’s measured failure mode was feature extrapolation: `Re_y` reached $1.85\times$ and $2.07\times$ the trained maximum on the two `Ret_360` ducts, and a gradient-boosted tree cannot extrapolate. A constitutive relation with a published constant *has no trained range*, so that mechanism cannot recur — and the two `Ret_360` ducts, exactly where the extrapolation cost lived, are the two that moved, by -0.0356 and -0.0375 . The one duct inside the old trained range, where round 4 was already close to the board’s best, is the one that gave ground.
2. **The pre-stated expectation band was wrong in our favour, and that is recorded rather than enjoyed.** The pre-registration’s band was $0.059\text{--}0.0666$ — rank 1 at its best end, a regression at its worst; scored after the fact, the rank the good end promised is worth 68% — 2–100% at 95% on eight cases — and its margin is **not statistically decided** (Table 2). The result landed at 0.056647, *below the good end of our own band*. What was nearly exact was the campaign’s plain arithmetic: matching the rank-2 entry’s published duct scores was computed in advance to give ≈ 0.056288 , against a measured 0.056647. The band was mis-centred pessimistic; the arithmetic underneath it was not (`CLOSURE_RANK1_CAMPAIN.md §1`).
3. **The cost was 37.0 solver core-min against a 95 core-min budget**, per-arm ledger in the pre-registration §7, with every QCR solve stopping on its own `residualControl` and no cap-stop anywhere in the route — a clause the freeze wrote in as route-voiding and never had to exercise.

3.3 What round 5 cost on AR_14_Ret_180, priced in writing and paid

The risk statement, and its outcome, in the order they happened. Before any test duct was solved, the rule freeze recorded that round 4’s 0.00003-level best-on-board lead on `AR_14_Ret_180` was *put at risk* by an all-or-none rule, named the training-side evidence pointing that way (`AR_10_Ret_180`, the largest-aspect-ratio training duct, had QCR’s weakest ratio at 0.664 and weakest correlation at 0.877, and `AR_14` extends that trend axis), and **accepted the possible loss in writing. The risk materialised.** `AR_14_Ret_180` went $0.0324698 \rightarrow 0.0353386$, $+0.0029$; the nominal tie with Reissmann’s published 0.0325 is **lost**; the case falls to 3 of 5, behind Reissmann and behind Wu & Zhang’s 0.0350; and our best-on-board count drops from 5 of 8 to **4 of 8**. The risk-side signal was right about which duct, and the comfort-side signal (`AR_7_Ret_180` at 0.477 and 0.928, far inside the bars) was right about the family.

It was not reverted, and the reason is the reason nothing else in this record is reverted: a post-hoc revert of one duct after seeing its score is per-case selection on test outcomes, which is what the benchmark’s one strict rule exists to prevent. The bundle is judged as a bundle. **The bundle wins by roughly twelve to one** — -0.0731 combined on the two `Ret_360` ducts

against +0.0029 here — and that ratio is the argument for all-or-none rules generally, not a reason to feel comfortable about this particular case. *We traded a nominal 0.00003 lead we had already told our own readers not to call a win, for 0.0088 on the overall. We would take that trade again; we would also have had to take it had the arithmetic gone the other way, which is the whole point of writing it down first.*

3.4 The per-case ledger

Table 4: Per-case scaled MAE across rounds, against the RANS-identity floor, the published rank-1 entry, and the best published score on each case. Lower is better. Round-5 values are the entry of record; round 2 is omitted for width and appears in §3.1. The overall row’s rank carries **P(rank 1) = 68%** (2–100% at 95%) and a lead over the published rank-1 entry that is **not statistically decided** (Table 2); the two Ret_360 rows are ties at the board’s published precision and are not wins.

Case	Floor	R3	R4	R5	Reissmann	Best pub.	Attribution
alpha_15_13929_4048	0.1320	0.0501	0.0501	0.0501	0.0592	0.0592	our model leads
alpha_15_13929_2024	0.2049	0.1011	0.1011	0.1011	0.1339	0.1195	our model leads
alpha_05_4071_4048	0.0461	0.0461	0.0461	0.0461	0.0606	0.0569	<i>baseline, declined</i>
alpha_05_4071_2024	0.0719	0.0719	0.0719	0.0719	0.0760	0.0760	<i>baseline, declined</i>
AR_1_Ret_360	0.1288	0.0919	0.0811	0.0455	0.0387	0.0387	2nd of 5; ties rank 2 at published precision
AR_3_Ret_360	0.1243	0.0862	0.0775	0.0400	0.0341	0.0341	3rd of 5, within published-precision noise of rank 2
AR_14_Ret_180	0.0590	0.0303	0.0325	0.0353	0.0325	0.0325	3rd of 5; the 0.00003 lead was priced and lost
NASA_2DWMH	0.0621	0.0632	0.0632	0.0632	0.0412	0.0364	behind, last on board
Overall	0.1036	0.0676	0.0654	0.056647	0.0595	—	rank 1 of 5, scored locally; P 68% (2–100% at 95%)

3.5 How the “best on board” count must be reported

Arithmetically we hold the best score on the public board on four of eight cases — five until round 5 spent one of them (§3.3). That count was on our public page as a headline and **we took it down ourselves**, because it is true arithmetic and misleading attribution (CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md §8.2). On alpha_05_4071_4048 and alpha_05_4071_2024 our submitted field *is* the organisers’ own unmodified RANS solve — re-derived independently from the supplied k - ω SST field and mesh and diffed against the shipped CSV to a maximum deviation of 5.0×10^{-10} , which is the CSV’s own write precision. Those two values are properties of a file every entrant is handed for free; anyone submitting it unchanged scores exactly the same. The honest attribution is Table 5.

Table 5: Attribution of the eight test cases at round 5, replacing the retired “5 of 8” headline. The “nominal lead only” row is empty because round 5 spent the case that occupied it.

Attribution	Count	Cases
Our model, clear lead	2	alpha_15_13929_4048, alpha_15_13929_2024
Our model, nominal lead only	0	— (was AR_14_Ret_180; lost in round 5)
The baseline led; the gate withheld our model	2	alpha_05_4071_4048, alpha_05_4071_2024
Behind	4	AR_1_Ret_360, AR_3_Ret_360, AR_14_Ret_180, NASA_2DWMH

The uncomfortable reading of that table is the one to state: **our overall number is now first on the board while our own model holds a clear lead on two cases out of eight.** The rank comes from having no bad case rather than from having many best ones — two declined baselines that beat every published entry, two ducts brought to the level of the entry whose constitutive term we adopted, and one case (NASA_2DWMH) where we are last and know it. An

unweighted mean over eight cases rewards exactly that shape, and a reader is entitled to know the shape and not only the mean.

What *is* defensible is a better finding than the one it replaces, and it is about the selection rather than the score: crossing the recorded per-case floor against the published per-case leader-board shows that **on exactly two of the eight test cases the uncorrected baseline beats all four published entries — and they are exactly the two cases the gate declined.** A rule fitted on 21 training cases and validated on 4 non-test cases, having never seen a test case, identified blind the two flows on which every published method in this field damages the answer. That is a finding about the benchmark which the steward may value more than our own number, and the draft cover email says so.

3.6 The same case, refused twice: a pattern rather than an incident

Round 4 improved two ducts and made a third worse: `AR_14_Ret_180` went from 0.0303 to 0.0325, +0.0022. It was not reverted, and the reason is the same reason the NASA hump’s +0.0011 was not reverted in round 2: choosing per case between two models *after* seeing their per-case test scores is selection on test outcomes, which is exactly what the benchmark’s one strict rule exists to prevent. Variant D was frozen on validation evidence and applied to all three ducts or none. It was applied to all three.

Round 5 put the identical choice to the laboratory a second time, on the identical case, with a bigger prize for cheating and a bigger loss for not: +0.0029 this time, and a best-on-board row spent rather than merely narrowed (§3.3). **The answer was the same, and it was written down before the number existed rather than defended after it.** A discipline that survives one test is a claim; a discipline that survives the same test twice, the second time at a higher price, is a practice.

Retired warning, kept visible rather than deleted. Earlier versions of this report carried a standing warning that `AR_14_Ret_180`’s best-on-board lead over Reissmann had collapsed from 0.0022 to **0.00003** (ours 0.0324698, against the published four-decimal 0.0325) and must never be reported as a comfortable win. That warning is now moot in the only way a margin warning ever becomes moot: **the margin was spent.** It is left in the record because a laboratory that quietly deletes its own cautions once they stop applying has taught its readers nothing about how it will treat the next one.

3.7 Three refusals

Three separate score improvements were identified, priced, and declined — each because taking it would have required choosing what to submit case by case from the answer key.

1. **0.0066, refused in round 2.** Switching the correction off on the three cases where round 2 was worse than doing nothing would have scored 0.0675. Knowing *which* three required reading round-2’s per-case test scores. The honest route — build a train-only gate first, validate it on non-test data, then apply it — recovered 0.0065 of the same term in round 3 and cost 0.0001. The laboratory’s own compliance audit calls this “the most defensible decision in the record”.
2. **The `AR_14_Ret_180` revert, refused in round 4** — worth 0.0022 on one case, 0.000275 on the mean — **and refused again in round 5**, at 0.0029 on the case and 0.00036 on the mean (§3.3).
3. **0.00014, refused in 2026-08-01 and still refused.** On `NASA_2DWMH` our correction is worse than not correcting (0.0632 against a floor of 0.0621); submitting the untouched solve there is one file copy, breaks no benchmark rule, and would move the overall from 0.056647 to 0.056510 — the arithmetic is unchanged by round 5, only its starting point

is. The only reason to single out that one case is that the benchmark told us its score. It is the smallest of the refusals and it was refused on identical grounds (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §8.3).

4 Where the deficit lived, and how the ducts closed it

4.1 The decomposition, and the mechanism it found

The C2 record decomposes the round-2 result case by case (`closure_challenge_C2_error_decomposition.md`, commit 1da013a8). Its sanity check ran first: recomputing the eight-case means from the per-case values reproduces the published floor 0.1036 and entry 0.0741 exactly.

Its headline was uncomfortable and is the reason the gate exists: **on three of eight cases our correction was worse than doing nothing**, and the two `alpha_05` regressions (+0.0262 and +0.0255) were an order of magnitude larger than the already known NASA hump regression (+0.0011), accounting for 98% of the 0.0528 of summed damage.

The first explanation offered — an “alpha-regime failure” — was replaced by the same document. Two facts kill it: the training split already contains five `alpha_05` cases, with unremarkable per-case training MAE of 0.0589–0.0719, so the model was not extrapolating into an unseen regime; and `NASA_2DWMH` is not a periodic hill at all yet sits in the same degraded group. The framing that survives is **the correction hurts where RANS was already good**: ordering the five PH-model cases by baseline quality separates them without overlap, hurting at every floor ≤ 0.0719 and helping at every floor ≥ 0.1320 , a gap of 0.0601 with nothing in it, Pearson $r = -0.9418$. *Caveat kept from the source: $n = 5$. A correlation of -0.94 on five points is suggestive, not established.* What earns it precedence over the alpha framing is not the r value but that it explains all three regressions with one mechanism, including the one the alpha story cannot touch.

4.2 Self-correction as a working practice

C2 is the most-corrected document in the closure line, and this report presents that as a feature rather than an embarrassment. Table 6 lists its corrections in place, each dated, each left visible in the record beside the text it corrects rather than silently rewritten.

The same practice runs through the rest of the record: a partial withdrawal of the sentence “no alternative training-family model exists for `NASA_2DWMH`” once CBFS was found shipped as legal training data (`closure_challenge_C6_hump_decision.md`, commit 1f5e2403); the retirement of the “5 of 8” headline; and a citation audit that found two of our own quotations had gone stale *because we improved the pages they cited* (§6).

4.3 The ducts: what was structurally missing

The deficit was the ducts, and this subsection is kept in the past tense it has earned rather than rewritten as though the diagnosis had always been obvious. Through round 4, the published rank-2 entry reached 0.0455 and 0.0399 on `AR_1` and `AR_3` where we reached 0.0811 and 0.0775; those two cases were 62.9% of the round-2 gap and remained the dominant term through four rounds. Four measured findings explain why, and **not one of them was fixable by more model capacity on the present inputs** — which is the finding that eventually forced the method change of §3.2 rather than another retraining.

1. **A proven algebraic degeneracy.** Uncorrected RANS produces *exactly* zero secondary flow on every duct: the streamwise fraction of mean $|\mathbf{U}|$ is 1.0000 on all four training cases, and the maximum transverse $|\mathbf{U}|$ is $\sim 1e-15$ against $O(10)$ streamwise, because a linear Boussinesq closure has no mechanism for Prandtl’s secondary flow of the second kind. For

Table 6: The C2 record’s corrections, all made in place and left visible. The docket describes C2 as “thrice-corrected”.

Date	What was corrected	What replaced it
in document	The “alpha-regime failure” framing	Baseline quality: the correction hurts where RANS was already good. Retained with its $n = 5$ caveat.
addendum	The floor was the wrong yardstick. Measuring against “do nothing” made the ducts look like wins (-28.6% , -30.7%)	Measured against the published board, AR_1 and AR_3 together are 62.9% of the gap to rank 2, and the two alpha_05 cases are the <i>smallest</i> recoverable terms (18.9% combined). The earlier recommendation to target alpha_05 is superseded by the document that made it.
2026-08-01 (ruling R6)	“Reproduce Wu, Zhang & Zhang’s β on the ω -destruction term”	DAFoam exposes betaFI0mega_ on the ω equation’s production term (DAk0megaSST.C:743); the destruction term’s β is the F1-blended model constant with no field hook. The two are not equivalent, because the other ω terms do not scale with β . What this stack can run today is a production-term inversion, and it is not a like-for-like reproduction.
2026-08-04 (W2 deep read)	One paper cited where there are two artifacts ; three claims did not survive reading both	(i) “train only on public CBFS” is true of the <i>entry</i> and false of the <i>paper</i> , which trains on mixed NASA hump + CBFS — and the hump is a scored case, so the paper’s protocol scored on this benchmark would be in-sample by construction; (ii) “zero-shot generalise to duct” is stated by neither artifact and is <i>our</i> inference from the entry’s training plus the leaderboard, sound but ours to carry; (iii) the rank-2 model is SST-QCRC , carrying an untrained QCR2000 quadratic constitutive term ($c_r = 0.3$) that no laboratory document had noticed.

a mean field of that form, I3_S3 and I4_W2S vanish identically — verified on the run’s own data at machine epsilon ($\sim 1e-14$) and independently over 50,000 random shear-rate pairs. Two of the seven features carry *zero* information on the entire duct family, by mathematical necessity.

2. **And it is worse than we published.** Two further identities were measured and had never been recorded: $I2_W2 = -I1_S2$ *bit-exactly* (residual 0.0 on all eight duct cases) and $I5_W2S2 = -I1_S2^2/2$ to 3.5×10^{-16} . **Five of the seven features carry one independent degree of freedom on a duct, not five**; the effective input dimension is 3, or 4 once Variant D’s $d/d_{\max}$ is added, where the physics is three-dimensional. The degeneracy is a property of the *baseline we are correcting*, not of duct geometry — the DNS field does carry secondary flow. Control: on the four periodic-hill test cases the same residuals run from 2.5×10^{-3} to 9.1×10^{-1} , so the test is not vacuous.
3. **A hypothesis of our own, tested and killed.** Every feature is an $O(3)$ invariant of $\nabla \mathbf{U}$ or a wall distance, all invariant under the $y \leftrightarrow z$ swap that maps the square duct’s quadrant onto itself — which looked like a hard error floor built in by construction. Measured on AR_1_Ret_180, the unreachable component is **0.012% of the correction’s energy**, because the true field respects the same symmetry the features do. Refuted, and shipped refuted.
4. **What binds instead is transfer.** A k -NN conditional-variance bound on the four training ducts puts the reachable error at 0.011–0.027 against floors of 0.069–0.107: in-family, the feature set is not the constraint. The binding constraint is the **Re_y** extrapolation of §3.1, and **no legal test of Reynolds transfer at a doubled velocity scale exists**

inside this benchmark — which is why Variant D is validated for no-harm on every held-out set the rules permit and is *not* proven to close the deficit.

Separately, and stated as a *bound* rather than a decomposition: the missing secondary flow is 2.07–2.22% of bulk velocity in the DNS and identically absent from RANS, so it can explain at most its own magnitude of our duct error — at most 24% on AR_1 and AR_3. **Scaled MAE does not decompose exactly, so this is a bound and not an error budget.** Its consequence is nonetheless directional: at least 76% of our duct error is streamwise-profile error, and rank 2’s *total* duct error (4.55%, 3.99%) is already smaller than our streamwise-only remainder ($\sim 7.0\%$, $\sim 6.6\%$). A scalar eddy-viscosity-style correction — exactly what the field-inversion line targets — is well aimed at the dominant term, not wasted against an anisotropy-only gap.

4.4 The measurement that named the problem: the amount was right, the structure was wrong

The finding that turned four rounds of duct diagnosis into a decision came from an evaluation census run on our own fields, not from the leaderboard (CLOSURE_EVALUATION_PROTOCOL.md §4.1; figures under `closure_eval/`). Its instrument is a streamline panel of the in-plane secondary flow, and its result is a single sentence: **our duct correction restores 69–112% of the true secondary-flow intensity while drawing the wrong topology.** On the square duct the LES field carries a counter-rotating corner-vortex pair — two distinct centres either side of the corner bisector, the canonical structure — and the corrected field draws **one** vortex. On AR_7_Ret_180 the correction reproduces the near-side-wall pair convincingly for $z/h > 5.5$ and replaces the truth’s rich multi-vortex region near the centreplane ($z/h < 1.5$) with a single smooth sweep.

Two things follow, and the second is the one that changed the plan.

- **An intensity scalar cannot tell those fields apart, and neither can the challenge metric**, which is a weaker scalar still. This is the concrete demonstration, on our own submitted family, that the score is satisfiable without the flow being right — a caution we would rather publish about ourselves than have a referee derive.
- **It is the a-posteriori confirmation of the a-priori degeneracy of §4.3.** Two of the seven features are provably zero on this entire family; the feature set is structurally short of what generating the correct vortex topology would require. *Asking a regression to synthesise, from features, a flow structure that the underlying solve does not contain at all is not a capacity problem.* That sentence is the whole argument for round 5, and it was measured before round 5 was proposed rather than assembled afterwards to justify it.

4.5 The falsifier: the duct zero is a property of the closure class

Before anything was applied to a test duct, the structural claim was given a falsifier and run against it on *training* cases only (`campaign/W3_QCR_DUCT_FALSIFIER.md`, 2026-08-05; the three scored test ducts were not opened). The pre-registered instrument was the toolchain’s own quadratic constitutive closure, run once on the smallest training duct’s existing mesh, with thresholds fixed in advance: in-plane RMS above 0.1% of bulk confirms the claim, below 1e-10 refutes it, between is indeterminate.

Neither OpenFOAM tree the laboratory holds ships QCR, so it was written: a ~ 180 -line library overriding `divDevRhoReff`, which reading the source showed to be *virtual on the turbulence model* rather than owned by the solver — so this is a turbulence-model-class change and no solver is patched or rebuilt. That fact is what made the route cheap, and it was measured rather than assumed.

AR_1_Ret_180: same mesh, BCs, forcing, schemes	in-plane RMS, % of U_b	converged
stock k0megaSST	3.16e-16	401 iters, residualControl
k0megaSSTQCR, $c_r = 0.3$	0.6239	401 iters, residualControl
reference LES/DNS field	0.7570	—

Verdict: the structural claim is confirmed, at six times the confirming threshold and 82% of the reference field’s own in-plane magnitude. Because a confirming direction is the weaker inference, it was strengthened twice rather than accepted: a $c_r = 0$ control on the same new library reproduces stock k0megaSST **bit for bit** ($\max |\Delta \mathbf{U}| = 0.0$), so the library adds nothing but the QCR term and the QCR term is the entire cause; and the recovered in-plane field correlates with the reference at Pearson $r = +0.953$, so it is the reference’s own vortex structure with the correct sign and not merely nonzero. Extended to all four training ducts the scaled MAE fell by 33.5–57.5%, from an untrained constant. Three caveats travel with those numbers and are kept: they are all-cell computations on training cases and are *not* challenge scores; AR_5 and AR_10 are cap-stopped rather than converged, so their absolute values carry an unquantified settling residue while the paired comparison stays fair; and QCR undershoots consistently, at 0.82–0.90 of the reference in-plane magnitude and never above it — the systematic slight weakness a trained coefficient would take up, and consistent with the published entry that carries this term also carrying a trained β beside it.

4.6 What round 5 drew

The census asked for topology, so topology is what the round-5 validation was checked against — on the benchmark’s own suggested validation duct, using the census’s own unmodified instrument, and recorded in the pre-registration as a *reported diagnostic and not a gate clause*, so that it could not quietly become a fourth criterion after the fact (`campaign/R5_PREREGISTRATION.md` §2).

The answer is yes, and it is the answer the post-hoc field could not give. QCR draws the counter-rotating corner-vortex pair with two distinct centres — one near $z/h \approx 6.1$, $y/h \approx 0.5$ and a second tight against the side wall near $z/h \approx 6.85$, $y/h \approx 0.55$ — in the same positions, with the same sense, and with the same flanking high-speed wall lobes as the LES panel. Mean in-plane magnitude is 4.65×10^{-3} of bulk against the truth’s 4.90×10^{-3} , and the visible deficit is stated rather than cropped: the LES’s weak tertiary recirculations near the centreplane ($z/h < 1.5$) are smoothed into elongated lobes. *The structure the ducts are scored on losing is the corner pair, and the untrained in-PDE term produces it where the trained post-hoc field, restoring the right amount of the wrong thing, could not.*

Independent corroboration nobody designed for. The published entry ranked second before round 5 — Wu & Zhang’s SST-QCRC — carries the same untrained QCR2000 term inside a larger model. Our three duct scores land **within 0.0004 of theirs on all three ducts**: 0.045470 against a published 0.0455, 0.039982 against 0.0399, 0.035339 against 0.0350. Different laboratories, different implementations, independent solves, the same constitutive term, the same answer to four decimals. Two readings are available and both belong in the record: it is a strong external check that our library implements the term correctly, *and* it says the duct signal is the QCR term itself rather than anything we added around it. We claim only the first as a credit to us.

4.7 What the two entries we passed do that we still do not

Heading changed 2026-08-07; the two readings below are unchanged. They were written when both entries were ahead of us on the overall. Between them they are still ahead of us on four of the eight cases, and a passed entry is not a refuted one.

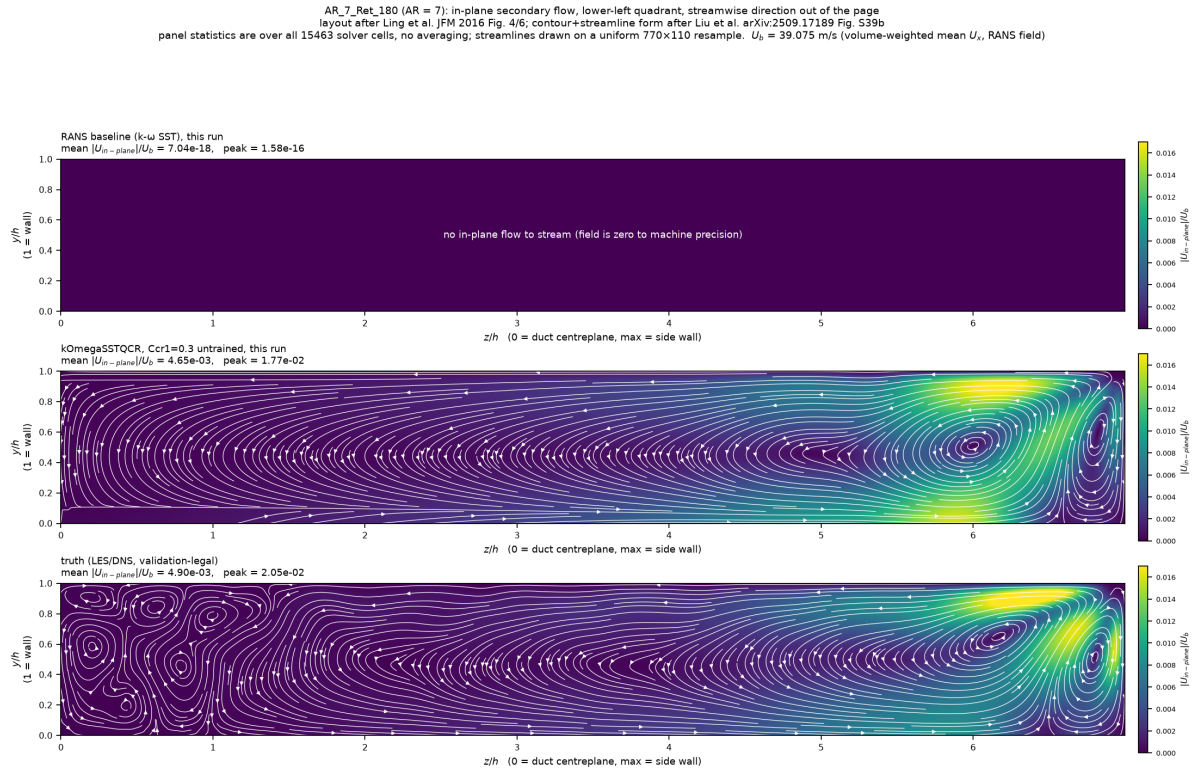


Figure 1: In-plane secondary flow on the validation duct AR_7_Ret_180, lower-left quadrant, streamwise direction out of the page; panel layout after Ling et al., *J. Fluid Mech.* (2016), per the instrument’s own header. **Top:** the k - ω SST baseline, which has no in-plane flow to stream — the field is zero to machine precision (mean 7.04×10^{-18} of U_b). **Middle:** the untrained kOmegaSSTQCR solve, mean 4.65×10^{-3} of U_b . **Bottom:** the LES truth, mean 4.90×10^{-3} . Panel statistics are over all 15,463 solver cells with no averaging. Source: `campaign/duct_secondary_flow_AR_7_validation_qcr.png`.

Versus Reissmann, Fang & Sandberg (0.0595; we are 0.0029 ahead on the overall and behind on all three ducts). Gene-expression-programming symbolic regression of an anisotropic Reynolds-stress correction, fitted concurrently with a differential-evolution refit of the k - ω SST coefficients (lineage: Weatheritt & Sandberg, *J. Comput. Phys.* 325 (2016) 22–37; implementation Reissmann, Fang, Ooi & Sandberg, *GPEM* 26(1):12 (2025), DOI 10.1007/s10710-025-09510-z). An algebraic stress correction inside the PDE *can generate* Prandtl’s secondary flow of the second kind and carries it across Reynolds number through the re-solve — exactly the two axes where they beat us worst. They train on the same duct family we do, AR 1–10 at $Re_\tau = 180$: **the difference is not data, it is model form**. Where we beat them: both `alpha_15` cases (0.0501/0.1011 against 0.0592/0.1339) and both declined `alpha_05` cases.

Versus Wu & Zhang (0.0624; we are 0.0058 ahead on the overall and level with them on the ducts to four decimals). One training flow, a five-constant symbolic law, and a re-solve beat our 21-case gradient-boosted model everywhere outside the periodic-hill family — and the way we closed that gap was to adopt half their model. Their SST-QCRC combines a fixed, untrained QCR-style quadratic Reynolds-stress term ($c_r = 0.3$, “directly adopted”, no data-driven training) with a learned multiplicative field β_{CND} on the ω -destruction term obtained by conditioned field inversion and PySR symbolic regression; the final law is three local features collapsed to $\beta_{\text{CND}} = \max(-0.1157, 0.0058525 \lambda_2) \cdot \lambda_2$, bounded below 4 by hand for stability (Wu & Zhang, *The Training of the SST-QCRC Model*, the entry’s own description document;

backing paper Wu, Zhang & Zhang, *AIAA J.* 63(2):687–706 (2025), DOI 10.2514/1.J064416, arXiv:2402.16355). What they do that we do not is **correct causes rather than symptoms** — the ω equation rather than the velocity field — and buy generalisation from compactness. The QCR term supplies duct secondary flow they never trained for. **What still separates us is the learned half**: they invert and regress a β_{CND} field on top of the constitutive term, and we ship the constitutive term alone. On the ducts that half is now worth, by measurement, at most 0.0004 — the whole distance between their published duct scores and ours (§4.6).

An external finding worth recording. Per their own paper’s main-text Table 1, the NASA wall-mounted hump — the same flow as the scored NASA_2DWMH — is one of the nine training flows of Liu, Wang, Zhao & Xiao’s method (arXiv:2509.17189). **Stated as an observation from their own table, not as an accusation**: the paper is not the submission, and whether the leaderboard entry retrained excluding the hump is not determinable from public sources.

5 The field-inversion line: blocked, unblocked, verified — and then negative

The decomposition points at a scalar eddy-viscosity-style correction inside the equations. That is the FIML family, and reproducing it is the laboratory’s Ladder-B line. Over the last fortnight it moved from a hard blocker to the first converged, verified closure-relevant gradient, and then to a negative result whose cause was found rather than guessed: **the inversion missed its pre-registered bar by two orders of magnitude, and the objective — not the adjoint, not the optimizer, not the term choice — was where the defect lived (§5.6–§5.8)**. It is the most useful section of this report for a reader trying to judge whether the laboratory’s successes are selected.

5.1 Capability established, and a deviation disclosed

Stage 1 (dafoam/ladder-b/S1_FIML_FIELD_INVERSION.md, commit 1cd44c04) established the capability on DAfoam v5.0.0 / OpenFOAM v2506 (He, Mader, Martins & Maki, *AIAA J.* (2020), J058853). Four results, in the order the evidence forces:

1. **The mesh-warp defect that fails this laboratory’s shape-optimisation cases cannot reach a β gradient** — measured, not reasoned. A counter monkeypatched onto DAfoamWarper.__init__ and compute_jacvec_product printed {"warper_init": 0, "warper_jacvec": 0} on every field-inversion adjoint that ran to completion. A class never instantiated cannot have a method called on it: the warp is not in the model. FIML is reachable on a code path where shape optimisation here is not.
2. **No custom turbulence library is needed**, and a previous record’s central disclosure was retracted on that basis: betaFI0mega_ and betaFIK_ are members of DAfoam’s own DAK0megaSST, constructed READ_IF_PRESENT with default 1.0.
3. **The disclosed deviation.** DAfoam’s β multiplies the ω equation’s **production** term; the destruction term’s β is the F1-blended model constant and carries no field hook. Anything built on this hook is a production-term inversion and must be recorded as such. **This is a real reduction in what Stage 1 can claim, and it is the honest one: building the production version and calling it the roadmap item would have been the quieter and worse choice.**
4. **The adjoint works and is finite-difference verified** on a 5,000-cell case with 5,000 design variables: 2.674% against the real objective gradient direction at step 10^{-3} and 2.683%

at 10^{-4} , with 2.2–3.1% per resolvable component and zero sign flips among components the finite difference can resolve. The residual $\sim 2.7\%$ is *step-independent* (a factor of ten in step moves it by 0.009 percentage points, where genuine central-difference truncation would fall by $\sim 100\times$) and *convergence-independent* (tightening the primal 1000-fold, to an achieved residual of $9.947\text{e-}12$, moved it from 2.6739% to 2.6728%). It is real, unexplained, and recorded rather than buried.

A finite-difference `check_totals` over a per-cell β field would need $2N$ primal solves — 10,000 on the tutorial case and 103,252 on the hump. Field-inversion gradients cannot be verified the way shape gradients are; the directional-plus-sampled-component protocol is the affordable substitute, and that is what “FD-verified” has to mean at this design-variable count.

A protocol defect found by auditing our own logs. A monitor replay flagged four of 34 finite-difference runs printing DA Foam’s velocity-bound message. The escalation asked which four points to exclude. **Both the question and its premise were wrong.** DA Foam gates that message on `printInterval`, which the campaign ran at the default of 100 — so each log reported clipping from 1% of its iterations. Re-running twelve points at `printInterval` 1 found **689 clip events against the 3 the archive recorded: a $230\times$ undercount**, with eleven of twelve runs clipping including the unperturbed baseline. There was no clip-free subset to retreat to. What the re-run also found is that all twelve objectives return *bit-identical to sixteen significant figures* with iteration counts matching exactly, and all 689 events fall in iterations 27–142 of runs 1815–3291 iterations long. The clip is a startup transient the iteration discards; the verification stands on a narrower and better-stated claim than before — not “no clip occurred”, which was never true and was never measured, but “the clip is a transient and the objective is provably unchanged by it”. The real defect found was in the protocol: the analyser read only the objective line and never inspected the log it came from.

5.2 The blocker, and its mechanism

Both closure-relevant target cases failed identically. The DA Foam/PETSc discrete-adjoint GMRES solve returned `PetscConvergedReason: -9 (KSP_DIVERGED_NANORINF)` at iteration 0 on CBFS (21,000 cells) and on the NASA hump (51,626 cells) — two meshes, two objectives, one signature. Memory was excluded by measurement (host `MemAvailable` bottomed at 18.99 GB with the Jacobian colouring completing normally), as were mesh quality, wall treatment, a non-differentiable velocity clip, primal convergence tolerance, and the objective function.

The mechanism was then located rather than guessed (`S1_FIML_FIELD_INVERSION.md` §4 addendum; `PROOF.md` §25.2–25.3):

- **The whole matrix-ordering axis was run.** Of DA Foam’s five orderings, two (`rcm`, `1wd`) produce the NaN and three (`natural`, `nd`, `qmd`) produce honest stagnation at -3 with the residual flat to 13 significant figures; *none* converges. The initial residual is identical ($7.091590452305\text{e-}04$) in all five, as it must be. A single buggy ordering would not do that; a factorisation hitting a destructive zero pivot under some elimination orders and not others does exactly that.
- **The cause was reproduced with no DA Foam in the loop.** The preconditioner matrix and right-hand side were dumped with stock PETSc runtime flags and analysed offline. The matrix (210,592 square, 13,710,468 nonzeros) has no zero rows, columns or diagonal entries. `scipy.sparse.linalg.spilu` returns “*Factor is exactly singular*” — the -9, in a second independent implementation — while `spilu`, full LU *with partial pivoting*, solves the same system to $\|Ax - b\|/\|b\| = 2.5 \times 10^{-12}$. The system is nonsingular and consistent. **Fill adds fill, not pivoting**, which is why raising the ILU fill level had never helped.

5.3 The unblock, and what it cost

`dafoam/ladder-b/W4_ADJOINT_PC_UNBLOCK.md` (commit 1cd44c04) closes it. The runtime-options route was tested first and found *reachable but empty*: `sub_`-prefixed factor options do land in the deployed factor (a $1e-8$ zero-pivot threshold is visible in `PCView`), correcting a standing belief that nothing reached the sub-preconditioner — and every one of them still returns `-9`, so the rebuild was justified by measurement rather than convenience.

The fix is one word. `DAFoam` hard-codes `PCType localPCType = PCILU` for the ASM sub-blocks; a single-hunk patch makes it `PCLU` when `DAFOAM_SUBPC_TYPE=1u` is set in the environment, and changes nothing otherwise. Results:

- **CBFS converges** — `PetscConvergedReason`: 2, 667 iterations, 243 s at 4 ranks — on B3’s exact `-9` configuration, changed only by the environment variable. The residual history is instructive: the first 400 iterations crawl, because the assembled matrix the LU inverts is not the matrix-free operator `GMRES` applies, and then the residual falls six decades in 267 iterations. **Slow-then-superlinear is what a correct strong preconditioner on a mismatched operator looks like; flat to thirteen digits is what the dead ILU looked like. They are not the same thing.**
- **The first closure-relevant field-inversion gradient exists**: 21,000 per-cell β components on CBFS, objective `varianceU` = $1.5279278906359758 \times 10^{-2}$, gradient norm 1.4558e-05.
- **The regression control was run before the treatment**: the patched image with the switch *unset* reproduces the `-9` at iteration 0 with all 13 residual digits identical. The rebuild is behaviour-neutral by default.
- **On the hump the same switch moves the failure from catastrophic to honest**: the factorisation completes and the residual descends monotonically ($1.094 \rightarrow 0.954$ over 900 iterations, accelerating) but does not converge, and the run was stopped deliberately when host `MemAvailable` fell to 1.62 GB on a shared box. **The run never produced a `KSPConvergedReason`**: it was killed by a `docker stop` this laboratory issued, not by an allocation failure or a solver verdict. So what is established on the hump is that the `-9` is gone and that *no convergence was observed within the 900 iterations run* — the convergence rate was never measured to completion and the memory envelope was never shown to be binding. **The hump adjoint boundary is uncharacterised**, and it is not claimed as a measured capability boundary in either direction: nothing here shows the operator is singular either.¹

Finite-difference verification of the CBFS gradient met its pre-declared gate at 13–45 times the required tightness (Table 7), with the unperturbed control run reproducing the adjoint run’s objective to *exactly zero difference* across all 17 digits. Total cost, billed at full occupancy for full wall-clock including a 600 s ordering timeout: 185.0 core-min against a 240 core-min budget.

The doubling of the error from $h = 0.05$ to $h = 0.1$ on cell 5491, with the finite difference moving *away* from the adjoint value as the step grows, is central-difference truncation behaving exactly as $O(h^2)$ predicts: the agreement is step-limited, not noise-limited, which is the signature of a correct derivative measured against a resolvable objective.

¹Corrected 2026-08-11 (hump-adjoint attempt audit). Earlier drafts of this report stated that “the hump blocker is no longer a singular factorisation — it is Krylov convergence rate against a memory envelope”. That is a causal claim resting on a single run killed at iteration 900 with no reason code, and it is withdrawn. Curing a singular ASM *sub-block* factorisation constrains the preconditioner, not the matrix-free operator `GMRES` applies. The measurements that would settle it are named and priced at `W4_ADJOINT_PC_UNBLOCK.md` §5b.1 (365 core-min, of which the first 40 are decisive), and they include an environment-off negative control that exists on CBFS and has never been run on the hump.

Table 7: Finite-difference verification of the CBFS β gradient (21,000 design variables). The last row was produced by an independent verification agent at a cell the laboratory never published.

Cell	h	Central FD	Adjoint $g[i]$	Rel. error	Source
5491 (largest $ g $)	0.05	1.914384790951e-06	1.916018813330e-06	0.085%	W4 §5d
5491	0.10	1.907246786675e-06	1.916018813330e-06	0.460%	W4 §5d
6740	0.05	1.679642395377e-06	1.678653382471e-06	0.059%	W4 §5d
12486	0.05	1.472401682783e-06	1.469472928907e-06	0.199%	W4 §5d
6490	0.05	1.520027166028e-06	1.519705840939e-06	0.021%	supervisor sweep

5.4 Independent adversarial verification

The claim was then attacked by a separate verification agent working under the doctrine that the claim is wrong until it survives attack (`dafoam/VERIFICATION_cbfs_unblock_supervisor_sweep.md`, commit `9c19ccc8`). Nothing was taken on the record’s word:

- The patch was diffed *against the built image*, not against the patch file. An md5 sweep of every `.C/.H` in the DAfoam source tree of both images shows exactly two changed files — `DALinearEqn.C` and its `lnInclude` copy — and all three AD-mode libraries contain the new environment-variable string, so all three were rebuilt from patched source.
- The finite-difference table was recomputed from the raw perturbation vectors: each perturbation file was verified to differ from the all-ones baseline at *exactly one index*, by exactly $\pm h$.
- A new finite-difference point was derived at cell 6490 — rank 10 of $|g|$, a cell the laboratory never chose — in an isolated case copy with the sweep’s own driver: 0.0211%, tighter than any published cell.
- The env-off regression was re-run cold, independently, reproducing -9 with all 13 residual digits.
- The cost ledger was reconciled against log timestamps to the second (693.04s measured against 693s billed; 642s summing exactly).

Verdict: no defect found, five of five attacks confirmed, one disclosed caveat restated rather than allowed to go quiet — the hump attempt of W4 §5b computes its objective against `NASA_2DWMH` experimental data, a scored case; it produced no gradient, nothing from it feeds any entry, and the gate claim rests on CBFS alone.

5.5 What the rank-2 method actually is, and what a reproduction must copy

The W2 reading (`campaign/W2_WU_ZHANG_DESTRUCTION_FIML_READING.md`, commit `393ad74f`) read both Wu artifacts in full and returned with the correction of Table 6 plus an implementation specification. Its operational consequences are three:

1. **A reproduction must copy the *entry’s* protocol, not the paper’s.** The paper’s models train on mixed NASA hump and CBFS inversion data; the hump is a scored case, so the paper’s protocol scored on this benchmark is in-sample by construction. The entry’s CBFS-only choice is what keeps a reproduction clean.
2. **A β -on-destruction patch alone reproduces the paper’s inversion capability, not the deployed rank-2 model.** Duct parity additionally needs the untrained QCR2000 constitutive term, filed separately as `w3-qcr-constitutive-term-for-rank2-parity`. Any score from a β -destruction-only model must not be set beside 0.0455/0.0399 as like-for-like.

3. The patch was specified from the installed source, not from documentation.

The ω equation is assembled in two places (DAkOmegaSST.C:745 in `calcResiduals()` and :870 in `getFvMatrixFields()`) and both carry the identical destruction line; patching only one would leave the preconditioner assembling a different equation than the residual — on a stack whose standing blocker *was* a factorisation of exactly that matrix, an inconsistent preconditioner would be a new confound indistinguishable from the old one. Seven named risks are recorded with the specification, including that the conditioned shielding function uses the constant 8 from the paper where OpenFOAM’s built-in DDES shield would silently supply 20.

5.6 The inversion that ran, and the gates it failed

The Stage 1 CBFS inversion was pre-registered (`dafoam/ladder-b/S1_CBFS_INVERSION_PREREGISTRATION.md`, commit `e6321e95`), written before any inversion iteration ran and while the only existing numbers were the baseline objective and the single verified gradient at $\beta = 1$. It fixed, in advance: the loss $J(\beta) = \lambda_{\text{QoI}} \text{varianceU}(\beta) + \lambda_{L2} \sum_j (\beta_j - 1)^2$ with $\lambda_{\text{QoI}} = 65.448114804931393$ so that the data term equals exactly 1 at $\beta = 1$ (Wu & Zhang’s own normalisation) and $\lambda_{L2} = 10^{-4}$ declared at the top of their stated band and *not* tuned; bounds $[0.2, 4.0]$, the upper bound theirs and the lower bound ours by choice and disclosed as such; L-BFGS-B rather than their SLSQP, because SciPy’s SLSQP would allocate a dense 3.749×10^9 -double workspace — 30.0 GB, the whole of this host’s RAM — at 21,000 design variables; a hard budget cap of 600 core-min with a stop rule that records a capped run as *capped*, *not converged*; and two success criteria, a 30% reduction in the data term (G1) and a requirement that more than half of the top-decile β deviations fall inside a pre-declared spatial window around the recirculation bubble and separated shear layer (G2).

It also fixed its own label before any result existed: **this is a production-term inversion**, with a field-variance loss rather than Wu & Zhang’s 30 sparse points — two independent reasons no number it produces may be set beside theirs as like-for-like.

It ran, and it failed both gates. 335.98 core-min against the 600 cap; 17 evaluations, every adjoint converged (reason 2, 667 iterations, every evaluation log); 10 accepted L-BFGS-B iterations with monotone descent; gradient norm down $106\times$ ($9.528\text{e-}4 \rightarrow 9.011\text{e-}6$); zero cells pinned at either bound. **G1: the normalised data term fell to 0.99851 against a ≤ 0.70 bar — a reduction of 0.149%, one two-hundredth of what was pre-registered. G2: 29.0% of the top-decile $|\beta - 1|$ cells sit in the declared window, against a $> 50\%$ bar** (`dafoam/ladder-b/S1_CBFS_INVERSION_RESULT.md`). Both are recorded as failed gates with no softening, and the stop is named for what it was: a decision at the plateau after 17 evaluations, not the optimizer’s own convergence test and not the budget cap.

The controls held throughout, which is what makes the failure interpretable rather than merely disappointing: evaluation 1 reproduced the W4 baseline objective to all 17 digits and the W4 archived gradient *bit-identically*, maximum absolute difference exactly zero; and a fresh cold process at the final β reproduced the final objective to all digits.

5.7 Why it failed: the objective, not the method

The post-run coverage audit found the cause, and it is a defect of ours in the case setup rather than a fact about turbulence closure. **The case’s inlet boundary condition was not the benchmark’s inlet.** Our O/U carried $U_x = 0.72$ uniform on all 150 inlet faces where the benchmark’s own CBFS case ships a developed profile ($0.202 \rightarrow 1.005$, bulk mean 0.9149). The mechanism is pinned rather than inferred: **the U_z noise columns of the two files are byte-identical**, so ours is the benchmark’s file with U_x and U_y overwritten — the leftover write-back of an earlier boundary-velocity pilot.

The consequence is arithmetic. The `varianceU` loss was comparing a 0.72-bulk RANS flow against a 0.915-bulk LES reference: **a 27% mass-flux mismatch that no interior turbulence correction can remove**. 85.1% of the total loss sat at $y > 2$, far above the step, in what should be trivially matched channel flow; the physics window the inversion exists to act in held 3.7%. Along the FD-verified gradient, a first-order 30% reduction would have required a step of norm ~ 315 in β -space, roughly 2.2 per cell, outside any physical bound. *The optimizer honestly minimised the loss it was given; the loss was measuring the inlet and not the closure. The 0.149% plateau is the honest optimum of the wrong objective.*

Three consequences are carried rather than filed away.

- **One sentence of W4’s record is corrected in place.** W4 §5c said the case’s “real nonuniform inlet is thereby restored rather than overridden”. The β design variable did stop overwriting the inlet each iteration, but the on-disk $0/U$ it inherited had already been overwritten; nonuniform-in- U_z -noise passed for nonuniform. A correction addendum is filed against W4. **W4’s own gate is unaffected:** finite-difference agreement is a property of the objective as built, and it stands.
- **A second real defect was bought on the way in.** DAfoam’s primal restarted in-process from its own converged state *walks away from it* (p residual $1e-6 \rightarrow 0.21$, ω and k pinned at their $1e-16$ floors), prints “Primal solution failed!”, and proceeds into the adjoint anyway, which then stagnates flat at reason -3 . In-process multi-evaluation optimisation is structurally unavailable on this case; the loop ran out-of-process, one fresh container per evaluation, at the same 16.4 core-min the cost model already stated.
- **Nothing from that β field may train anything.** It is an artifact of a defective loss. Stage 2 is blocked pending the repair, and is recorded as blocked rather than as pending.

5.8 The re-inversion, in flight

The repair was pre-registered before any solver run of it, on a fresh case copy with the corrupted case left untouched as evidence, and with a falsifiable prediction attached to the diagnosis so that the diagnosis could be wrong in public (`dafoam/ladder-b/S1_CBFS_REINVERSION_PREREGISTRATION.md`, commit `a11ccd41`): *P1 — the loss floor collapses, baseline varianceU falls from $1.528e-2$ to below $7.6e-3$; if it is still $\sim 1.4e-2$ the inlet diagnosis was wrong and that is the headline of the item.*

Stages A–C are measured, at 96.57 core-min of a 450 core-min hard cap, and the inversion itself has not yet started:

- **P1 passes beyond its prediction.** The repaired baseline is **$6.1509e-04$** , a **$24.8\times$ collapse** against a prediction of merely “more than half”. Inlet-adjacent bulk is now 0.9148 against the LES 0.9153, ratio 1.0005: the mass-flux floor is gone.
- **P2 passes.** The $y > 2$ free-channel share of the loss fell from 85.1% to 25.6%, and the physics window rose from 3.7% to **41.2%** while holding 8.4% of the cells — the largest loss-per-cell concentration in the domain, which is where a closure correction should be asked to act.
- **A protocol change was forced by a measurement that first looked like a failure.** The re-verification finite differences missed their own 1% bar at 25.9–32.2% on three cells. The diagnosis was that the repaired case’s cold primal now starts far from its solution and stops at the first $1e-6$ crossing, leaving $O(\text{tol})$ state error that under-develops a one-cell perturbation’s response. The discriminating experiment — the same finite-difference pair at $1e-8$ — moved that cell from 25.9% to **0.032%**. *The adjoint was right; the tolerance was polluting the finite difference.* Every solver run of the item now runs at $1e-8$, and the re-anchored sweep is 0.032%, 0.115% and 0.009% on three cells in distinct mesh neighbourhoods, zero sign flips, all inside W4’s own order of agreement.

- **The gate numbers were fixed by dated amendment before the driver launches**, re-based to the repaired baseline rather than inherited: $\lambda_{\text{QoI}} = 1.6258 \times 10^3$, G1 at $\text{varianceU} \leq 4.3056 \times 10^{-4}$, G2’s window confirmed against the stage-A audit rather than assumed, and λ_{L2} cut tenfold to 10^{-5} with the reasoning disclosed — the failed run plateaued at an exact penalty balance ($|g_{\text{pen}}|/|g_{\text{QoI}}| = 0.998$, $\cos = 0.9995$), which the repaired gradient, $8.4\times$ larger, should not reproduce.

The honest summary of this line is therefore neither the triumph of §5.3 nor the failure of §5.6 alone: **a verified gradient was used to minimise a loss, the minimisation worked, and the loss turned out to be measuring our own broken boundary condition — which the pre-registered gates caught, and which nothing else in the run would have.** A gate that only ever passes would have let this ship.

6 Verification and integrity

6.1 Leakage guards, in code rather than in prose

The one strict rule was audited line by line in the code rather than taken from our own documentation (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §4.1):

- Ground-truth reads occur only inside loops over the 21 training cases. The test-case loop calls the RANS-field loader only, annotated `# no U_LES read`.
- The split is enforced by executable assertions, not comments: `assert len(_PH_TRAIN) == 21 and len(_PH_VAL) == 4`, and assertions that the training, validation and test name sets do not intersect. The test list is imported *solely* to assert non-intersection.
- The CSV exporter does not merely omit scoring calls: before any pipeline work it replaces `score`, `score_from_csv`, `evaluate_by_case`, `evaluate_from_csv_by_case`, `evaluate_individual_case` and the ground-truth accessors with raising stubs across three namespaces, then *proves the guard is armed* by calling `score()` and catching the refusal before continuing. If a future edit reintroduces a scoring call, the script dies rather than quietly spending one.
- An in-sample gate (`closure_in_sample_gate.py`) runs before and after closure work and passes.

The residual soft leakage, disclosed by us before anyone asks. Across rounds, test feedback influenced *which machinery got built*. The round-3 record states it in its own words: *“The gate’s MOTIVATION came from round 2’s official call . . . That is soft, adaptive leakage, and it is real.”* The defence is that the gate’s every parameter derives from 21 training cases and was reproduced byte-for-byte from train-only data before touching test features; that seeing round-2’s scores changed *when* it was applied, not *what* it outputs, so its four test decisions would have been identical had round 2 never been scored; and that the benchmark itself ships the ground truth and instructs submitters to preview per-case scores, so an interpretation under which observing them is a violation would make the benchmark’s own documented workflow a violation. **Verdict: not a violation of the rule as written; residual risk low but non-zero, and interpretive rather than factual.** It belongs in the submission’s cover material so that a reviewer *finds* it written down rather than discovers it.

Round 5’s leakage surface, and why it is the smallest the entry has ever carried. The soft leakage above is a statement about *motivation*, and round 5’s motivation was as adaptive as any: we knew our three duct scores. What round 5 removes is the channel through which

knowing them could act. Nothing in the three duct predictions is fitted to anything, so there is no parameter that could have been fitted on the wrong data; the test cases were opened for RANS inputs and mesh only, and the *_LES files were never copied into the run tree, so test truth was not merely unread but **absent from the working directory**; and the one degree of freedom through which known scores could have entered — choosing which ducts get which model — was removed in advance by an all-or-none rule, which is the subject of the next subsection.

6.2 The frozen rule, and a result nobody had to accept

Round 5 is the round in which the laboratory’s pre-registration discipline was asked to do the harder of its two jobs. Saying no to a candidate that fails its own gate costs nothing but the candidate. Saying *yes* — accepting a number mechanically, including its losses, with no discretion left to exercise — is the half that decides whether the discipline is real, and it is the half a laboratory only gets to demonstrate when the result is good.

The order of events, which is the whole claim.

1. **2026-08-05T17:41Z — the application rule is frozen and committed (R5_RULE_FREEZE.md at 0bade54a), before the first validation iteration existed.** QCR ships on all three test ducts if and only if, on the benchmark’s own suggested validation duct AR_7_Ret_180: **V1** QCR’s scaled MAE is $\leq 0.70 \times \text{SST’s}$; **V2** the in-plane Pearson r is ≥ 0.85 ; **V3** both arms converge on their own residualControl within the cap. Otherwise none. The all-or-none form was chosen precisely because per-case selection is the one place known test scores could enter.
2. **The validation duct is solved and the clauses are read off it.** V1: 0.0400 against 0.0839, ratio **0.4770**. V2: **0.9284**. V3: both arms stopped by residualControl at 4,652 and 4,660 iterations against a 12,000 cap. **Verdict ALL. No clause was added, dropped, or re-thresholded after the freeze** (campaign/R5_validation_AR7.json).
3. **The accept criterion is frozen before the scoring call (e865076b):** the submission improves on the entry of record *iff* overall < 0.065438 , with the expectation band 0.059–0.0666 and the AR_14 risk statement written down in the same document.
4. **Four independent pre-score checks are run fresh by the scoring agent**, not trusted from the producing session: the five unchanged CSVs SHA-256-match the round-4 manifest; all eight CSVs are 1000×3 and finite; the evaluation-point ordering check is re-run from scratch, maximum row-wise deviation $4.9e-8$; and the pre-registration’s own hash table matches the bytes on disk for all eight files. All four passed, and only then did the call proceed.
5. **The call returns 0.056647, which is less than 0.065438, so the criterion says ACCEPT.** Round 5 became the entry of record by arithmetic.

No judgement was exercised after the number existed. The acceptance, the non-revert of AR_14, and the reporting of a lost best-on-board row were all determined by documents committed before the number was produced — by a different agent than the one that produced it.

Two honest qualifications belong beside that box, in the same spirit as the ones beside the gate.

- **A frozen rule is only as good as the evidence it acts on, and ours acted on one validation case.** AR_7_Ret_180 is the benchmark’s own suggested duct validation case and it is a single case at $Re_\tau = 180$. **No legal duct truth exists anywhere in this benchmark at $Re_\tau = 360$** , so the transfer to the two Ret_360 test ducts was defended structurally — an untrained term has no trained range — and not empirically. That argument was written down before the run, and “cannot fail the old way” was explicitly not claimed to be “cannot fail”.
- **The band was wrong and the rule was not.** A pre-registration whose expectation band misses low is a pre-registration whose author did not know the answer, which is the

condition under which freezing rules is worth anything. We record the miss as a miss (§3.2) rather than presenting 0.056647 as the number we expected.

6.3 The scoring-call ledger

Six distinct prediction sets have ever been scored against the eight test cases’ ground truth: the RANS-identity floor, and rounds 1 through 5. The sixth call was made on 2026-08-07 by the supervisor’s designated scoring agent — not by the session that built round 5, which stubbed every scoring and truth entry point of the evaluation package and verified itself armed before touching the first case, and which stated in its pre-registration that the call belonged to somebody else. The unit counted is *distinct prediction sets*, which is a defensible and meaningful unit — and the compliance audit caught a docstring claiming something stronger and demonstrably false (“exactly one call” where the same run makes four invocations across two prediction sets, the extra pair re-scoring the already-counted floor as a harness check). The claim was substantively harmless and was fixed anyway, because **an entry whose entire credibility rests on precise self-accounting cannot ship with a sentence a reviewer can falsify by reading forty lines further down the same file.**

The ledger is a self-imposed discipline. The benchmark imposes no limit and instructs submitters to preview their score. No public statement of ours may imply otherwise. Nor may any statement of ours imply the ledger is principled: in Blum & Hardt’s own words (arXiv:1502.04585, §2.3 above) rate limiting is a “poorly understood heuristic”, and naming ours as one is the honest price of it.

6.4 The adversarial methods audit

An audit conducted on 2026-08-04 proof-checked the closure line end to end under the instruction to assume the records overstate until verified, test-blind throughout (CLOSURE_METHODS_COMPARISON.md, commit ac2f37ee). The metric definition held as written. Six gaps were found in our own evidence, ranked by how much the entry’s credibility rests on the unverified assumption, each costed as a proposal:

- G1 — seed sensitivity, real and live.** `random_state=0` everywhere and no model ever re-trained under another seed. This is not pedantry: at 327,600 training cells the PH model crosses scikit-learn 1.9.0’s 200,000-row binning subsample, so the bin edges — hence the trees, hence three of the eight predictions — depend on the seed, against a gap to rank 2 of 0.0030.
- G2 — no physicality check on any corrected field.** $\nabla \cdot \mathbf{U}$ never measured, on the one dimension every other entrant gets by construction. **Closed twice:** as a measurement on 2026-08-05, returning a material finding against us, and as a *defect* for the duct family on 2026-08-07, when round 5 replaced those fields with solves (§6.5).
- G3 — baseline convergence evidence exists but is uncited** for all eight predictions. It matters twice over on the two declined cases, where the shipped solve *is* our submission. **Closed 2026-08-05** (CLOSURE_CHALLENGE_STATUS.md §4b, commit 11ff63b1): the four periodic-hill test cases carry final residuals of $\sim 1\text{e-}10$ after 20,000 iterations, and a sweep of all 29 cases reproduces the pattern on 28 of 29 — with the exception recorded honestly rather than smoothed, one *training* case whose shipped residual file is truncated at iteration 76, so that its convergence is plausible but not evidenced. The three duct test cases terminate with “*SIMPLE solution converged*” at 405 / 1540 / 7009 iterations under explicit `residualControl`, and the $O(0.4)$ – $O(0.7)$ transverse residuals that look alarming are recorded as a normalisation artifact on a machine-zero field ($\max|U_y| = 2.1 \times 10^{-15}$ against U_x up to 108 m/s), noted so that nobody later mistakes them for a divergent solve.

G4 — manifest gap. The three new round-4 duct CSVs carried no recorded SHA-256 and the round-4 submission directory had no manifest; the hashing discipline covered only the five copied files. **Closed 2026-08-05** (`closure_challenge_submission_round4/MANIFEST.json`, commit 30fd6d7a): all eight files hashed, zero scoring calls, guaranteed by a script that never imports the benchmark package at all.

G5 — gate statistical power is thin and already at its disclosed limit; widen from 4 held-out decisions to 25 by leave-one-out over all periodic-hill cases.

G6 — no training-convergence curve for either gradient-boosted model.

None requires a scoring call; none touches test ground truth; all six keep the leakage guards armed. Four are now closed: G1 and G2 were taken first, on pre-registered thresholds (§6.5), and G3 and G4 landed on 2026-08-05.

6.5 Seed stability and field physicality: the lines drawn before the measurements

The pre-registration for G1 and G2 was committed (`closure_challenge_stability_physicality_audit.md` §0, commit 73fa6a33; results at `fee5eaf9` and `cbe040f6`) *before* either run existed, so the materiality lines cannot have been drawn around the results:

- **G1:** retrain the PH model at eight seeds with everything else frozen, score on the four validation cases only, and compute a test-side prediction-spread bound requiring no ground truth at all. *Material if* the overall-equivalent induced spread reaches 0.0003 (10% of the gap to rank 2) by either estimate, *or* if any single case’s spread reaches 0.0030 — a one-case spread the size of the whole gap being material regardless of the mean.
- **G2:** push both the raw RANS and the corrected field through the *identical* already-validated Green–Gauss operator so that discretisation error cancels in the comparison, and report the volume-weighted RMS divergence and its ratio. *A ratio of 2 or more on any model-corrected case is recorded as a material physicality cost of the method class and must be carried into the submission’s disclosure section; a ratio below 2 is still reported per case, quantified, with no editorialising either way.*

G1, measured. The results are recorded in §1 of the same audit file (commit `fee5eaf9`), with raw numbers in `closure_challenge_seed_sensitivity.json` (generated 2026-08-04T18:55Z; zero scoring calls; guard re-armed and proven armed first; ground truth read for the 21 training and 4 validation cases only). Seed 0 reproduces the recorded round-1 model exactly (validation pooled 0.0876) and its predictions match the shipped round-4 CSVs to 5e-10. Across seeds 0–7 the pooled validation metric spans only 0.000276, and the two proxy ranges for the served `alpha_15` predictions are 0.000568 and 0.000520; the overall-equivalent proxy estimate is $S_{\text{proxy}} = 0.000261$, just below the 0.0003 line. But the test-side spread *bounds* — the largest disagreement between any two seeds’ own predictions at the 1000 evaluation points, normalised the same way the metric is — reach **0.0085** and **0.0099** on the two `alpha_15` cases and 0.0010 on the hump, giving $S_{\text{bound}} = 0.00242$.

Verdict: material, by two of the three pre-declared triggers — $S_{\text{bound}} \geq 0.0003$, and two single-case bounds at or above 0.0030 — and both halves of the reading belong in the record:

- **Where truth is legitimately available, the metric is stable under seed.** On the four held-out validation cases the per-case spread is 0.0005–0.0012 and the overall-equivalent is ~ 0.00026 , a tenth of the gap to rank 2. That is the best available *estimate* of the seed effect on the score, and it is small: the entry gains a measured rather than assumed stability property, an estimated one-seed uncertainty on the overall of about 0.0003.

- **Where truth is not available, the only honest statement is the bound, and it is not small.** Across seeds the trees move the pointwise test predictions by 0.85–0.99% of the velocity scale on the two served `alpha_15` cases — per-case movement up to roughly three times the entire gap *if it failed to cancel against the truth*. On validation cases that wobble demonstrably cancels to the 0.0006 level in the metric; expecting the same on test is reasonable, but it is an extrapolation and not a measurement, and the pre-registered rule counts the bound. Tightening it would cost a scoring call and is not taken.

The consequence was fixed in advance and has been executed: **our leaderboard-gap language now carries a seed-uncertainty qualifier** in the status record and on the public page. Scope is exact — the duct model behind three predictions is bin-deterministic at 41,971 cells, so `random_state` is inert for it by construction rather than by measurement, and the two declined cases are the organisers’ own solve, so the five non-PH predictions contribute zero seed variance and the division by eight is exact.

G2, measured — and the finding is against us. The divergence audit completed on 2026-08-05 (310 s on the two-core cap, zero scoring calls, 37 cases; results in §2 of the audit file, raw record `closure_challenge_divergence_audit.json`). Every regenerated test field was matched against its shipped round-4 CSV *before* its divergence number was believed — maximum absolute difference 4.8e-10 to 4.9e-8, all at CSV write precision — so the numbers are about the submitted fields and not a lookalike.

Table 8: Volume-weighted RMS divergence of the eight submitted fields, both fields through the identical validated Green–Gauss operator. The scale-free column is the continuity error relative to the field’s own velocity-gradient magnitude, $\|\nabla \cdot \mathbf{U}\|/\|\nabla \mathbf{U}\|_F$, given for the RANS field and for the submitted field; the RANS value is the operator-consistency floor, since the solver enforces continuity on face fluxes rather than on reconstructed cell gradients.

Case	Submitted field	Ratio	rel. RANS	rel. corrected
<code>alpha_15_13929_4048</code>	PH-corrected	58.0	0.18%	10.5%
<code>alpha_15_13929_2024</code>	PH-corrected	123.6	0.08%	9.7%
<code>alpha_05_4071_4048</code>	declined (raw RANS)	1.000	0.47%	0.47%
<code>alpha_05_4071_2024</code>	declined (raw RANS)	1.000	0.27%	0.27%
<code>AR_1_Ret_360</code>	duct-D corrected	$\sim 1\text{e}15^\dagger$	2.4e-17	2.3%
<code>AR_3_Ret_360</code>	duct-D corrected	$\sim 4\text{e}14^\dagger$	6.9e-17	3.1%
<code>AR_14_Ret_180</code>	duct-D corrected	$\sim 7\text{e}14^\dagger$	4.6e-17	3.4%
<code>NASA_2DWMH</code>	PH-corrected	1.526	0.43%	0.66%

[†] The duct ratios are divisions by machine zero and should be read as such: a fully developed, exactly unidirectional RANS duct field is cell-wise divergence-free to roundoff (9e-12 against a gradient scale of 4×10^5). The interpretable statement is the last column, not the ratio.

Verdict: material. Thirty-five of thirty-six model-corrected cases sit at or above the pre-registered ratio of 2, and five of the six corrected submissions exceed it; only `NASA_2DWMH` (1.53) stays under, for the consistent reason that the PH-trained correction is small relative to that case’s own gradient scale — the same model that barely moved its score barely moves its continuity error. The train and validation cases show the same corrected-field levels (8–19% on hills, 2–3% on ducts), so this is a property of the method on its own turf and not a test-set artifact.

Three things follow, and none of them is an excuse:

- **The cost is real and now has a number.** The correction takes fields whose measured continuity error sits at the operator floor and returns fields violating continuity at $\sim 10\%$

of the gradient scale on the two corrected hills and 2.3–3.4% on the three ducts. That is far above the operator floor, so it is the field and not the operator.

- **The scoring metric never sees it.** Nothing measured here changed round 4’s 0.0654. But it is exactly what a referee of the description document will ask, and per the pre-registered consequence it is carried into the submission’s disclosure section — “we measured it and here is the number” beats silence, even when the number is against us.
- **The two declined cases are, again, the strongest rows.** Their ratio is exactly 1.000 because the gate’s “off” state ships the organisers’ own solve and inherits its physicality untouched. The one part of our submission that is physical by construction is the part where our model does not appear.

G2, closed for the duct family — by changing the method, not the threshold. Round 5’s three duct predictions are the output of a converged SIMPLE solve, in which continuity is enforced by the pressure equation on the face fluxes rather than asserted afterwards. Pushed through the *identical* Green–Gauss operator the G2 audit used (`closure_challenge_round5_qcr_forward.json`; `campaign/R5_PREREGISTRATION.md §6`):

Case	round-4 ML field, RMS $\nabla \cdot \mathbf{U}$	round-5 QCR field	improvement
AR_1_Ret_360	9,160	345	27×
AR_3_Ret_360	9,749	169	58×
AR_14_Ret_180	3,584	57	63×

Stated at the precision it deserves rather than rounded into a slogan: the *cell-centred* Green–Gauss divergence of the QCR fields is **not** the baseline’s machine zero, because SIMPLE enforces continuity on face fluxes and the explicit QCR stress divergence enters the cell-to-face (Rhie–Chow) consistency at the 5e-4-of-gradient level. What is true is narrower and sufficient: a 27–63× reduction on the identical operator against the fields it replaces, and a residual non-solenoidality that is **solver-consistency small rather than model-structure large**. G2 is closed for the duct family and remains open, measured and disclosed for the two corrected hills at ~10%.

This whole finding is also the measured case for the field-inversion line of §5, and round 5 is the first instalment of it. The argument for moving the correction *inside* the equations had been an argument from taxonomy and from the duct decomposition; the G2 audit made it an argument from a number on our own submitted fields; and round 5 acted on it with the cheapest in-PDE term available. A correction that re-enters the PDE and is re-solved returns a field satisfying continuity because the solver enforces it — which is precisely what every other entry on this board gets for free, what we now get on three cases, and what Stage 1 is being built to give us on the rest.

6.6 The pre-registration that said no

The clearest evidence that the discipline can refuse is the `alpha_05` regime candidate — filed under an “R5” name it never earned, because it was never scored and never became a round (`closure_challenge_R5_ALPHA05_REGIME_PREREGISTRATION.md`, commit `be642415`). C2 had recommended training a second, regime-specific model as strictly better than gating, because it recovers the term instead of merely declining to lose it. The candidate was built — 11 of 21 training cases qualified under a membership rule defined by the same frozen C1 gate that makes the test-time decision, and the gate refit reproduced $\alpha = 0.7499$ and threshold 0.1263 before membership was computed, the script exiting rather than defining membership from a drifted gate.

The go/no-go rules were declared in the committed training script’s docstring before any leave-one-out delta existed. Measured:

Table 9: The `alpha_05` regime candidate’s pre-declared gate, and what the measurements did to it.

Rule	Measured	Verdict
(i) the validation case improves	<code>alpha_05_10071_4048</code> : 0.0759 $\rightarrow$ 0.0447 (−0.0312), where the global round-1 model gives 0.0772, a hurt	PASS
(ii) mean leave-one-out $\Delta < 0$	−0.0195 over 10 cases, 8 of 10 improved, two by more than 0.035	PASS
(iii) no case hurt by more than +0.010	<code>alpha_05_4071_3036</code> : 0.0492 $\rightarrow$ 0.0659, +0.0167	FAIL

Verdict: NO-GO. The rules were not revisited after the numbers, the scoring call was not authorised, and nothing was scored. Round 4 stood as the entry of record, and the duct route of §3.2 — not this one — eventually became round 5. Read beside §6.2, the pair is the point: the same machinery refused a candidate it had built and accepted one it had priced pessimistically, and in neither case did anyone choose after the fact.

The rule has a name, and ours is the weaker version of it (added 2026-08-05). “Adopt the retrained model only if it is certified not to be worse than the incumbent” is a named research problem: Thomas, Theocharous & Ghavamzadeh, *High Confidence Policy Improvement*, *Proc. 32nd ICML*, PMLR 37 (2015) 2380–2388 (abstract read 2026-08-05 from the PMLR record). Their algorithm gives “probabilistic guarantees about the quality of each policy that it proposes”, with the user free to “select any performance lower-bound and confidence level.” Two differences cut against us and one for us. Against: their bound is probabilistic and user-parameterised, where **our +0.010 per-case hurt cap is a single hand-chosen number whose pre-registration never states a derivation** — it was declared before the Δ s existed, which is the part that matters for leakage, but “why 0.010” is unanswered in our own record, and the method-priority review (`CLOSURE_METHOD_PRIORITY_REVIEW.md` §2.4, §7, commit `d84b649f`) files that as a charter trigger with a proposal, `agenda/proposals/w8-a-hurt-cap-that-states-where-it-came-from.json`. Also against: clinical trials pre-specify a **non-inferiority margin** in the protocol for exactly the reason we froze ours, and our cap is that idea under another name rather than bespoke machinery. For us: their guarantee is about the aggregate, ours is per-case, which is stricter — and it is why R5 failed on one case while improving on eight of ten.

The finding underneath is worth more than the candidate would have been (recorded as lesson L-33). The case that fails is `alpha_05_4071_3036` — the training sibling of both declined test cases, sharing their α and their 4071 parameter and differing only in the third index — and it is the lowest-baseline case in the entire training split (0.0492). The leave-one-out table splits cleanly on baseline quality: every case with baseline ≥ 0.058 improves and both cases below ~ 0.055 get worse, in both variants. **C2’s mechanism is recursive: train only on the low-baseline regime and the hurt-where-RANS-is-already-good boundary does not disappear — it moves down, from ~ 0.07 – 0.13 to ~ 0.055 , and reappears inside the regime.** A correction model fitted to a population still damages the best-baseline members of that population.

A successor candidate — a second-stage threshold declining even the regime model below ~ 0.055 — is obvious and is explicitly *not* adopted, because it was conceived after those numbers existed and adopting it now would be exactly the rules-revisited-after-the-fact move the document’s own §3 forbids. It is filed as a recommendation needing its own pre-registration, its own membership audit, and a held-out design that does not lean on a single in-regime validation case.

6.7 Corrections and withdrawals register

- **The published duct diagnosis was wrong** and was corrected by the round that acted on it (§3.1).
- **“No alternative training-family model exists for NASA_2DWMH” was withdrawn:** CBFS — the curved backward-facing step, the nearest flow in this benchmark to a wall-mounted hump — ships as *training* data with its LES fields and is legal to fit on. The pre-registered argument that chose the periodic-hill model was a two-way comparison of hills against ducts in which CBFS is not named, not rejected and not mentioned: the choice was defended against the worse of two candidates while the better third sat unused in the training set. What still keeps the case shut is not the absence of a candidate but *who* may choose and *when* — anyone choosing today has seen round 2’s score, so taking that route legally requires a pre-registration frozen first. Both routes are priced; neither has been taken.
- **“We are last on the ducts” was never true.** A working belief repeated in internal notes was re-derived from the harness and found false; after round 4 we are last on NASA_2DWMH alone.
- **Two of our own quotations went stale because we improved the pages they cited.** A citation audit extracted all 28 quotations in the submission draft mechanically and matched them against normalised source text. Both defects were in citations of sources *we control* — not one was in a citation of the benchmark’s rules, which had been transcribed carefully because they were known to matter. The audit closes the question that no sentence in that document is attributed to a source nobody here has read, and explicitly does *not* close the general problem: nothing checks staleness automatically, so it must be re-run at the moment of approval, not before.
- **An over-generalisation in our own solver record was corrected on measurement:** a prior finding that a certain PETSc failure code fools DAFoam’s success gate was found to be specific to -5; -9 never fools it. It matters in the safe direction — the blocked cases were never at risk of silently returning a wrong gradient.
- **“The only entry on the board that does not re-solve” was retired on 2026-08-07,** not softened. It became false of three of our eight predictions the moment round 5 shipped, and the self-description at the front of §2.1 now leads with that rather than qualifying it in a footnote.
- **“4 of 8” replaced “5 of 8”** on the best-on-board count, by losing a row we had already told our readers not to count as a win (§3.3). The retired margin warning is left in place rather than deleted.
- **One sentence of the W4 unblock record was found false by our own follow-up audit and corrected in place:** the field-inversion case’s inlet had been overwritten by an earlier pilot, so the claim that its “real nonuniform inlet is thereby restored” was wrong, and the inversion built on it minimised the wrong objective (§5.7). The defect was ours, was found by us, and was found by an audit run *after* the run had already failed its gates — which is the order in which such things are usually not found.

7 Outlook

7.1 The technical path

The duct term is spent, and it was the largest one. Through round 4 the two `Ret_360` ducts were 1.8 times the entire deficit to the published rank-1 entry, in one flow family. Round 5 took them, and what is left of that family is 0.0068 and 0.0059 against Reissmann on `AR_1` and `AR_3` plus the 0.0029 given back on `AR_14`. **Nothing else on the board is now worth what the ducts were**, and the next-largest single term is the hump at 0.0220 on one case, which §4 has already established cannot be legitimately attacked.

Full rank-2 parity is now a smaller target than it looked. The remaining measured target is the other half of the model we half-adopted. The deployed SST-QCRC is the untrained QCR term *plus* a conditioned β on ω destruction. We now ship the first half and match their duct scores to 0.0004, which prices the second half on this family at essentially nothing (§4.6). The β -on-destruction model patch remains specified line by line against installed source and filed — but it is filed as a *physics* question about term choice (§5.7), not as a score route, and no score from it may be compared like-for-like with theirs.

Stage 1 has reported, and the honest label is: machinery yes, result no. The CBFS production-term inversion ran to 335.98 of its 600 core-min cap and failed both pre-registered gates because its objective was measuring an overwritten inlet. The repair is pre-registered with a falsifiable prediction, the prediction passed at 24.8 $\times$, the finite-difference gate is re-anchored on the repaired objective at 0.009–0.115%, and the re-inversion is in flight against a 450 core-min cap (§5.8). **What it can establish is unchanged and still modest**; what it has already established is that this laboratory’s gates catch its own defects.

The hump remains open, and its adjoint blocker is uncharacterised. The one attempt that got past the -9 was killed at iteration 900 with no `KSPConvergedReason`, so we can say no convergence was observed in the iterations run, and we cannot say whether the limit is convergence rate, memory, or the conditioning of the operator itself — corrected 2026-08-11, having previously been stated as “Krylov convergence rate against a memory envelope rather than a singular factorisation”. A Richardson-wrapped configuration capping the restart basis is staged and unrun, and is one of six measurements priced at 365 core-min that would convert the assumption into a measurement. Its *entry* blocker is different and unmoved: no legal held-out flow sits at the hump’s Reynolds number, the nearest donor extrapolates on 7 of 7 features, and we already know the case’s score — so any model choice made today is made by somebody who has seen it. One untried probe would answer the physics question legally (QCR forward on CBFS, a training case), and it is correctly priced at 78 core-min rather than the 10 an earlier draft claimed.

Audit items. Four of the six are closed: G1 and G2 both returned material findings against pre-registered thresholds, G3 and G4 landed on 2026-08-05, and G2 was then closed a second time for the duct family by round 5. What remains is G5, which widens the gate’s evidence base from 4 held-out decisions to 25 by leave-one-out over all periodic-hill cases, and G6, which is free if folded into the retraining runs. **Both remaining items are about the periodic-hill model** — which, after round 5, is the only learned object left in the entry.

7.2 Submission status

The package exists, is audited, and now carries the stronger entry. The eight CSVs are written, 1000 rows by 3 columns, comma-delimited and header-free, verified finite and round-trip accurate

below $5e-8$, with mean velocity magnitudes falling inside the band spanned by all four accepted submissions on all eight cases — a check that would catch a point-ordering, column-ordering or units blunder at zero scoring cost. All eight round-5 files are hashed and cross-asserted, five of them against round 4’s own manifest. Both defects the compliance audit found (the false docstring; the non-existent submittable artifact) are closed, the evaluation package is cited by commit hash rather than by its misleading version string, and the flagged preprint quotations have been verified verbatim.

The description document’s disclosures are fixed and non-negotiable, and round 5 changed what several of them say rather than whether they appear: that the entry is now two method classes and which cases belong to which; that two submitted fields are the uncorrected baseline and any leaderboard-best status there belongs to the baseline; that the duct predictions carry an untrained published constant and nothing fitted; the full adaptive-leakage history in our own words; the executable train/validation/test assertions; that the hump regression was not reverted and why, and that the `AR_14_Ret_180` regression was not reverted either and was priced in writing first; the duct feature degeneracy as a negative result we would rather publish than have inferred from our scores; the scoring-call count, now six, correctly framed as self-imposed; the measured continuity cost of §6.5 on the five post-hoc predictions and its $27\text{--}63\times$ improvement on the three duct predictions; the seed-uncertainty qualifier on every gap statement; the prior art, split as §2.2 splits it into classifiers that *identify* and classifiers that *control*, with no sentence presenting confidence-gated correction as novel; the Hanna et al. citation for the post-hoc method class, with its delta stated (§2.1); and Spalart’s QCR2000 as the source of the duct term, adopted untrained and credited plainly rather than described as ours.

And the package is not being sent today. What remains is not technical: the author line and reference URL, and the proofread and approval without which nothing moves. Two questions are drafted for the steward and unsent — whether a company name is acceptable on a leaderboard row, and whether any terms of use apply to a commercial entity training on data whose repositories carry no licence file at all. **Submission is held at the owner’s discretion.** That is a status, not a verdict: the package is complete and audited, nothing about it is blocked, and no part of this report should be read as saying the work was abandoned or the route closed. The number stands, dated, scored locally, with every caveat this document attaches to it; the decision to put it in front of a steward is Katie’s, and it is hers to take at whatever moment she judges right.

A Provenance index

Table 10: Repository artifacts cited in this report. Paths are relative to `demo-output/website/` unless otherwise shown; commits are those at which the cited text stands.

Artifact	Commit	What it carries
<code>CLOSURE_CHALLENGE_STATUS.md</code>	<code>07a7fe9e</code> (§0f), <code>b9cb56a2</code> (§4)	Rounds 1–5 history; the metric; per-case tables; feature-expressivity and generalisation studies; §0f is the round-5 entry of record
<code>CLOSURE_METHODS_COMPARISON.md</code>	<code>ac2f37ee</code>	Metric read from the evaluation code; entrant dossiers; the dimension-by-dimension comparison; gap findings G1–G6

Artifact	Commit	What it carries
closure_challenge_C2_error_decomposition.md	1da013a8	The error decomposition and its corrections
CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md	15530f97	Policy, eligibility, compliance audit, draft package, addenda
CLOSURE_CHALLENGE_PRIOR_ART.md	15530f97	Prior art on classifier-gated corrections: §2.1 identify, §2.3–§2.4 control, §2.7 the PMoE near miss read in full
CLOSURE_METHOD_PRIORITY_REVIEW.md	d84b649f	Priority and novelty review: 27 searches, 13 counted negatives, the eight owed citations and the two phrasing corrections applied in §2.1–§2.3, §6.3 and §6.6
closure_challenge_C1_error_estimator.md	5719374e	The decline gate: fit, validation, and its stated statistical limits
closure_challenge_C6_hump_decision.md	1f5e2403	Both hump routes priced; neither taken
closure_challenge_R5_ALPHA05_REGIME_PREREGISTRATION.md	be642415	The round-5 candidate and its NO-GO
closure_challenge_stability_physicality_audit.md	73fa6a33 (§0), cbe040f6 (results)	G1/G2 thresholds committed before either run; results §1–§2
closure_challenge_seed_sensitivity.json	cbe040f6	G1 raw record: 8 seeds, spreads, bounds, verdict
closure_challenge_divergence_audit.json	cbe040f6	G2 raw record: 37 cases, RMS divergence before and after correction
closure_challenge_submission_round4/MANIFEST.json	30fd6d7a	All eight submission CSVs hashed; closes G4
closure_challenge_round5_qcr.json	07a7fe9e	The entry of record: the sixth scoring call, its per-case table, the four pre-score checks, the AR_14 risk statement scored against its outcome
campaign/R5_RULE_FREEZE.md	0bade54a	The all-or-none application rule, frozen before the first validation iteration existed
campaign/R5_PREREGISTRATION.md	e865076b	Round 5: gate receipts, per-case hashes, risks, accept criterion, topology check, continuity table, cost ledger
campaign/R5_validation_AR7.json	e865076b	V1/V2/V3 measured on the validation duct
closure_challenge_round5_qcr_forward.json	e865076b	Forward-solve machine record; QCR divergence through the G2 operator
campaign/duct_secondary_flow_AR_7_validation_qcr.png	e865076b	Figure 1: SST, QCR and LES in-plane streamlines

Artifact	Commit	What it carries
<code>campaign/W3_QCR_DUCT_FALSIFIER.md</code>	303247bb	The QCR library, the falsifier verdict, the $c_r = 0$ bit-exact control, the four-duct battery
<code>CLOSURE_RANK1_CAMPAIN.md</code>	0afd7e61	The costed routes, the duct arithmetic, and the de-ranking of the learned- β route on its own measurement
<code>CLOSURE_EVALUATION_PROTOCOL.md</code>	3a05ca70	The metric census: structure-versus-amount, score-versus-physicality
<code>closure_challenge_trained_entry_round4_duct.json</code>	4c2f3562	The round-4 entry, superseded
<code>closure_challenge_duct_reynolds_transfer.json</code>	d0992da2	The round-4 pre-registration
<code>dafoam/ladder-b/S1_FIML_FIELD_INVERSION.md</code>	1cd44c04	FIML capability, FD verification, the clip audit, the blocker
<code>dafoam/ladder-b/W4_ADJOINT_PC_UNBLOCK.md</code>	1cd44c04	The sub-block LU unblock and the first closure-relevant gradient
<code>dafoam/VERIFICATION_cbfs_unblock_supervisor_sweep.md</code>	9c19ccc8	Independent adversarial verification of that claim
<code>dafoam/ladder-b/S1_CBFS_INVERSION_PREREGISTRATION.md</code>	e6321e95	The first inversion, pre-registered before it ran
<code>dafoam/ladder-b/S1_CBFS_INVERSION_RESULT.md</code>	c3903781	Both gates failed; the inlet-corruption diagnosis; the primal-restart defect
<code>dafoam/ladder-b/S1_CBFS_REINVERSION_PREREGISTRATION.md</code>	a11ccd41	The repair, its graded predictions, the $1e-8$ protocol change, and the gate numbers fixed before launch
<code>campaign/W2_WU_ZHANG_DESTRUCTION_FIML_READING.md</code>	393ad74f	Both Wu artifacts read in full; the patch specification
<code>campaign/PROBABILITY_OF_RANK_2026-08-10.md</code>	b032bbae	Table 2: the 400,000-draw case-level bootstrap, the four intervals in order of increasing honesty (Monte Carlo, leave-one-out, double bootstrap at 68% and at 95%), and the head banner withdrawing the internal-only gate
<code>campaign/LADDER_V_PASS3_COLD_2026-08-11.md</code>	9477a2ed	The cold reproduction: §3 (W2) is the independent $P(\text{rank } 1) = 0.674$ from public data, the finding that withdrew the gate

Artifact	Commit	What it carries
<code>campaign/LADDER_V_TRIPLE_VERIFICATION.md</code>	5bffb476	Ladder V; V8 and its 2026-08-10 amendment — every rank claim carries the probability, its interval and the not-decided pairs

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